



Sukhoi Su-35

The **Sukhoi Su-35** (Russian: Сухой Су-35; NATO reporting name: **Flanker-E/M**, occasionally nicknamed "**Super Flanker**"^[6]) is the designation for two improved derivatives of the Su-27 air-defence fighter. They are single-seat, twin-engine, supermaneuverable, 4.5 generation air superiority fighters, designed by the Sukhoi Design Bureau and built by Sukhoi.

The type was originally developed by the Soviet Union from the Su-27 and was known as the **Su-27M**. It incorporated canards and a multi-function radar giving it multi-role capabilities. The first prototype made its maiden flight in June 1988. Following the dissolution of the Soviet Union, Sukhoi relabeled it **Su-35** in hopes of attracting export orders. Fourteen aircraft were produced and used for tests and demonstrations; one example had thrust-vectoring engines and was in turn redesignated the Su-37. A sole Su-35UB two-seat trainer was also built in the late 1990s that resembled the Su-30MK family.

In 2003, Sukhoi embarked on a second "deep" modernization of the Su-27 to serve as an interim export aircraft awaiting the development of the Sukhoi PAK FA (Su-57) program. Also known as the Su-35, this version incorporates technology from the PAK FA program and has a redesigned cockpit and weapons-control system and features thrust-vectoring engines in place of the canards.^[7] The type made its first flight in February 2008. Although it was designed for export, the Russian Air Force became the launch customer in 2009, with the production version designated Su-35S. China's People's Liberation Army Air Force has also placed orders.

Su-27M / Su-35



A Russian Aerospace Forces Su-35S

General information

Type	Multirole fighter
National origin	Soviet Union (Russia since 1991)
Manufacturer	United Aircraft Corporation (for Su-35S)
Designer	Sukhoi Design Bureau
Built by	Sukhoi
Status	In service
Primary users	Russian Aerospace Forces People's Liberation Army Air Force Algerian Air Force
Number built	Su-27M: 12 ^[1] Su-35S: >151 ^{[2][3][4][5]}

History

Manufactured	Su-27M: 1987–1995 Su-35S: 2007–present
Introduction date	February 2014
First flight	Su-27M: 28 June 1988 Su-35S: 19 February 2008
Developed from	Sukhoi Su-27
Variant	Sukhoi Su-37

Design and development

Upgraded Su-27

The first aircraft design to receive the Su-35 designation had its origins in the early 1980s, at a time when the Su-27 was being introduced into service with the Soviet Armed Forces. The definitive production version of the Su-27, which had the factory code of T-10S, started mass ("serial") production with the Komsomolsk-on-Amur Aircraft Production Association (KnAAPO) in 1983. The following year, this Su-27 version reached initial operational readiness with the Soviet Air Defence Forces.^[8] Having begun work on an upgraded Su-27 variant in 1982,^[9] the Sukhoi Design Bureau was instructed in December 1983 by the Soviet Council of Ministers to use the Su-27 as the basis for the development of the Su-27M (T-10M).^[10] Nikolay Nikitin would lead the design effort throughout much of the project's existence, under the oversight of General Director Mikhail Simonov, who had been the chief designer of the Su-27^[11] along with Mikhail Pogosyan.



Planform view of the Su-27M

While sharing broadly the blended wing-body design of the Su-27, the Su-27M is visibly distinguished from the basic version by the addition of canards, which are small lifting surfaces, ahead of the wings. First tested in 1985 using an experimental aircraft,^[8] the canards, in complement with the reshaped wing leading-edge extension, redirected the airflow in such a way so as to eliminate buffeting at high angles of attack and allowed the airframe to sustain 10-*g* manoeuvres (as opposed to 9 *g* on the Su-27) without additional structural reinforcement.^[12] More importantly, when working with the relaxed-stability design and the accompanying fly-by-wire flight-control system, the aerodynamic layout improved the aircraft's manoeuvrability and enabled it to briefly fly with its nose past the vertical while maintaining forward momentum. Because of this, theoretically, during combat the pilot could pitch the Su-27M up 120 degrees in under two seconds and fire missiles at the target.^[13] Other notable visible changes compared to the T-10S design included taller vertical tails, provisions for in-flight refuelling and the use of two-wheel nose undercarriage to support the heavier airframe.^{[14][15]}

Besides the increase in manoeuvrability, another feature that distinguished the Su-27M from the original design was the new weapons-control system. The centrepiece of this system was the multi-function N011 Bars (literally "Leopard") phased-array radar with pulse-Doppler tracking that allowed it to detect targets below the horizon. First installed on the third prototype, the radar transformed the Su-27M from simply being an air-defence fighter into a multi-role aircraft capable of attacking ground targets.^{[8][16]} Compared to the N001 Myech ("Sword") radar of the Su-27, which could track 10 targets and only direct two missiles towards one target at a time, the new radar could track fifteen targets and direct missiles towards six of them simultaneously.^[8] The extra weight of the N011 radar at the front of the aircraft necessitated the addition of the canards; engineers would only later discover the aerodynamic advantages of these devices.^{[1][17]} In addition, an N012 self-defence radar was housed in the rearward-projecting tail boom,

making the aircraft the first in the world to have such a radar.^[16] Other changes to the aircraft included the use of uprated turbofan engines, as well as the increased use of lightweight composites and aluminium-lithium alloys in the aircraft's structure.^{[14][18]}

Testing and demonstration

In 1987, Sukhoi started converting the first prototype (designated T10M-1) from a T-10S airframe at its experimental plant in Moscow. Although it had canards, the first prototype, like several subsequent aircraft, lacked the many physical alterations of the new design.^[19] It made its first flight after conversion on 28 June 1988, piloted by Oleg Tsoi, followed by the second prototype in January 1989.^[20] Following the conversions of the two Su-27M prototypes, the actual production of the aircraft was transferred to the country's Far East where it was carried out by KnAAPO. The third aircraft (T10M-3), which was the first new-built Su-27M and first to be constructed by KnAAPO, made its first flight in April 1992.^[20] By then, the Soviet Union had disintegrated, and the ensuing economic crisis in Russia throughout the 1990s meant that the original plan to mass-produce the aircraft between 1996 and 2005 was abandoned,^[8] with the aircraft to serve as experimental test-beds to validate the canards, the flight-control system and thrust-vectoring technology.^[14] In total, two prototypes, nine flying pre-production and three production aircraft were constructed by 1995;^{[1][21]} the production aircraft were delivered in 1996 to the Russian Air Force for weapons testing.^[22]



The ninth Su-27M in-flight

By the time of the disintegration of the Soviet Union, Sukhoi had been demonstrating the Su-27M to senior defence and government officials. With its debut to a Western audience at the 1992 Farnborough Airshow, the company redesignated the aircraft as Su-35.^[23] The aircraft subsequently made flying demonstrations overseas in an effort to attract export orders, starting in November 1993 with Dubai, where Viktor Pugachev flew it in a mock aerial engagement with an Su-30MK in front of spectators.^{[24][25]} The aircraft then flew in Berlin and Paris, and would be a regular feature at Moscow's MAKS Air Show.^[25] The Russian government cleared the

aircraft for export during Sukhoi's unsuccessful sales campaign in South Korea during the late 1990s and early 2000s;^[26] the company also marketed the aircraft to Brazil, China and the United Arab Emirates.^[27]

As the flight-test programme of the Su-27M proceeded, engineers discovered that the pilot failed to maintain active control of the aircraft during certain manoeuvres, such as the Pugachev's Cobra. The eleventh Su-27M (T10M-11) was therefore equipped with thrust-vectoring engine nozzles in 1995, and the resultant Su-37 technology demonstrator made its first flight on 2 April 1996.^{[28][29]} It also tested the enhanced N011M radar, as did the twelfth developmental Su-27M.^[27] The Su-37's ability to maintain a high angle of attack while flying at close to zero airspeed attracted considerable press attention.^[30] It later received different engines and updated fly-by-wire controls and cockpit systems for evaluation.^[27]

Apart from the single-seat design, a two-seat aircraft was also constructed. Working in cooperation with Sukhoi, KnAAPO's own engineers designed the Su-35UB so as to combine thrust-vectoring engines with features of the Su-27M. Modified from an Su-30MKK airframe, the aircraft made its first flight on 7 August 2000, and afterwards served as an avionics test-bed.^[31] While the original Su-27M never entered mass production due to a lack of funding,^[32] Sukhoi refined the Su-27M's use of canards and the Su-37's

thrust-vectoring technology and later applied them to the Su-30MKI two-seat fighter for the Indian Air Force.^[33] The tenth Su-27M (T10M-10) also served as a test-bed for the Saturn AL-41F1 engine that is intended for the Sukhoi Su-57 (previously known under the acronym "PAK FA") jet fighter.^[34]

Modernization

With the need to update Russia's aging fleet of Su-27 aircraft, Sukhoi and KnAAPO in 2002 started integrating glass cockpits and improved weapons-control systems (to accommodate a greater variety of weapons) to air force aircraft. The Su-27SM, as the modified aircraft is called, made its first flight in December 2002.^[35] The success of this project led Sukhoi in December 2003 to proceed with a follow-up "deep" modernization programme. Known internally as T-10BM,^[1] the programme was aimed at a more thorough redesign of the airframe to narrow the qualitative gap between Russian aircraft and foreign fourth-generation aircraft. The resultant design, also designated Su-35, would serve as an interim solution pending the introduction of the PAK FA fifth-generation fighter (which eventually became the Su-57), many features of which the aircraft would incorporate.^{[1][36][37]} The aircraft was primarily aimed at the export market, being offered by KnAAPO as a single-seat alternative to the two-seat Su-30MK variants built by Irkutsk Aviation Plant (IAP).^{[38][39]}



Thrust vectoring nozzle on a Sukhoi Su-35S

In many respects, the T-10BM design resembles the Su-27 more than the Su-27M. During tests of the thrust-vectoring engines and the Su-27M's aerodynamic layout, Sukhoi had concluded that the loss of manoeuvrability due to the removal of the canards – the addition of which imposed a weight penalty on the airframe – could be compensated for by the addition of thrust-vectoring nozzles.^[N 1] Industry progress in the fields of avionics and radars has also reduced the weight and size of such components, which shifts the centre of gravity of an aircraft rearward.^[41] Accordingly we saw SW, designers removed the canards (and the dorsal air brake) found on the Su-27M; the size of the vertical tails, aft-cockpit hump and tail boom were also reduced.^[41] With such changes, as well as the increased use of aluminium and titanium alloys and composites, designers had reduced the empty weight of the aircraft, while maintaining a similar maximum take-off weight to the Su-27M.^{[42][43]}



Su-35S cockpit layout: a head-up display, two multi-function liquid crystal displays, and a control stick with HOTAS controls

While the Su-27M design had the avionics to give the aircraft the nominal designation as a multi-role fighter, flight tests with the Russian Air Force revealed difficulties in deploying the aircraft's armament efficiently. According to *Aviation Week & Space Technology*, air force pilots described weapons trials with the aircraft in Akhtubinsk and Lipetsk as a "negative experience", with a particular emphasis on the layout of the cockpit and its adverse impact on the workload of the single pilot.^[40] Designers, test pilots and avionics software specialists therefore worked together to redesign the cockpit and its attendant systems and improve the human-machine interface. The information management system of aircraft's avionics suite had been changed so that it now has

two digital computers which process information from the flight- and weapons-control systems. The information is then displayed on two 9 in × 12 in (23 cm × 30 cm) multi-function liquid crystal displays,^[44] which replaced the smaller multi-function cathode-ray tube displays found on the Su-27M.^[40] The pilot can also view critical flight information on the head-up display,^[45] and is equipped with Hands On Throttle-And-Stick (HOTAS) controls.^[44]

The Su-35 employs the N035 Irbis-E ("Snow Leopard") passive electronically scanned array (PESA) radar, which is a further development of the N011M radar that had been evaluated on Su-27M test-beds and constitutes the core of the Su-35's weapons-control system. It is capable of detecting an aerial target up to 400 km (250 mi; 220 nmi) away, and can track thirty airborne targets and engage eight of them simultaneously; in addition, the multi-function radar is capable of providing high-resolution images of the ground using synthetic aperture mode.^[46] The aircraft is equipped with an OLS-35 optoelectronic targeting system ahead of the cockpit to provide other forms of tracking including infra-red search and track.^[46] For defences against enemy tracking, the Su-35 is equipped with the L175M Khibiny-M electronic countermeasure system,^[47] while engineers have applied radar-absorbent materials to the engine inlets and front stages of the engine compressor to halve the Su-35's frontal radar cross-section and minimise the detection range of enemy radars.^[48] The radar cross section of the Su-35 is reportedly 1 to 3 square meters.^[49] The multi-role Su-35 can deploy air-to-air missiles of up to 300-kilometre (190 mi) range, and can carry the heavy Oniks anti-ship cruise missile, as well as a multitude of air-to-ground weaponry.^{[50][51]}

The Su-35 is powered by a pair of Saturn AL-41F1S turbofan engines, formerly known as *izdeliye* (Product) 117S. A highly upgraded variant of the AL-31F, the AL-41F1S is closely related to the Su-57's Saturn AL-41F1 (*izdeliye* 117), differing primarily in the engine control system.^{[53][54]} The engines are equipped with thrust-vectoring nozzles, which have their rotational axes canted at an angle; the nozzles operate in one plane for pitch, but the canting allows the aircraft to produce both roll and yaw by vectoring each engine nozzle differently; this configuration was first implemented on the Su-30MKI and is also used on the Su-57.^[55] The Su-35's thrust-vectoring system and integrated flight- and propulsion-control systems allow the aircraft to attain 9-*g* manoeuvres and "supermaneuverability", enabling it to perform post-stall manoeuvres at low speeds. This differs from Western air combat doctrine, which emphasises the maintenance of a fighter aircraft's kinetic energy.^[52] The engine gives the Su-35 the limited ability to sustain supersonic speed above Mach 1.1 without the use of afterburners, also known as "supercruise".^[42]

"The classical air combat starts at high speed, but if you miss on the first shot—and the probability is there because there are maneuvers to avoid missiles—the combat will be more prolonged. After maneuvering, the aircraft will be at a lower speed, but both aircraft may be in a position where they cannot shoot. But supermaneuverability allows an aircraft to turn within three seconds and take another shot."^[52]

— Sergey Bogdan, Sukhoi chief test pilot

Testing and production

Following the completion of design work, KnAAPO constructed the first prototype, which was finished in mid-2007. The prototype, Su-35-1, was then transferred to the Gromov Flight Research Institute at Zhukovsky Airfield, in preparation for its maiden flight.^{[56][57]} On 19 February 2008, Sergey Bogdan took the aircraft aloft for its 55-minute first flight from Zhukovsky.^{[56][58]} Bogdan later piloted the second prototype on its maiden flight on 2 October from KnAAPO's Dzyomgi Airport.^[59] The flight-test

programme was expected to involve three flying prototypes, but on 26 April 2009, a day before its scheduled maiden flight, the fourth Su-35 (there's a static test aircraft) crashed during a taxi run at Dzyomgi Airport. The aircraft struck a barrier at the end of the runway and was destroyed by fire; the pilot ejected and sustained burn injuries.^{[60][61]} The cause of the accident was the failure of the engine management system, as a result of which the afterburner was not turned off.^{[62][63][64]}



A Russian Aerospace Forces Su-35S during a standard takeoff procedure

Although the Su-35 project was aimed primarily at the export market, the actual launch order for 48 Su-35S aircraft was placed by the Russian Defence Ministry at the 2009 MAKS Air Show (as part of a larger deal worth US\$2.5 billion for 64 fighter aircraft).^{[6][65][38]} During the early stages of the flight-test programme, Sukhoi estimated that there was such a market for 160 aircraft, with a particular emphasis on Latin America, Southeast Asia and the Middle East. Some of the candidate countries, such as Algeria, Malaysia, and India, were already operators of the Su-30MK family. As the aircraft was to be available for export starting in 2010,^[66] During the type's international debut at the 2013 Paris Air Show, Mikhail Pogosyan, General Director of Sukhoi's parent company United Aircraft Corporation, stated that there was an estimated demand for 200 aircraft, split evenly between the domestic and export markets.^[67] It was not until the end of 2015 when the first export contract was signed with China; by then, the Russian government had placed a follow-up order for 50 aircraft.^[68]

Apart from the launch order at the 2009 MAKS Air Show, the Russian government and the state-owned Vnesheconombank development bank agreed to provide Sukhoi with capital for the aircraft's production. In November 2009, KnAAPO (which was renamed KnAAZ in 2013 after it became part of the Sukhoi Company) started manufacturing the first production aircraft,^[69] the general assembly of which was completed in October 2010;^[70] by then, pilots and engineers had successfully completed preliminary tests of the aircraft's systems.^[71] The first Su-35S took its maiden flight in May 2011, and would be delivered (along with other aircraft) to Akhtubinsk to start state joint tests with the Defence Ministry to prepare the aircraft for service. Because production of the Su-35S occurred alongside trials, some early-production aircraft were later modified as a result of such tests.^[72] In December 2018, United Aircraft Corporation has reported 100th serial Su-35S was produced at the Komsomolsk-on-Amur Aircraft Plant.^[73]

Operational history

Russia

In 1996, three production Su-27Ms were delivered to the air force's 929th State Flight Test Centre named for V. P. Chkalov (GLITs) at Vladimirovka air base, Akhtubinsk, to perform weapons trials.^[22] In 2001, the air force decided to transfer several Su-27Ms to re-equip the Russian Knights aerobatics team, and so the team's pilots took familiarisation flights with the aircraft.^[74] The three production and two other pre-production Su-27Ms arrived at the team's Kubinka air base near Moscow in 2003. However, they were used as a source of spare parts for other aircraft in the demonstration fleet.^[75]



A Russian Air Force Sukhoi Su-35 performing at MAKS 2009 air show in Moscow

Initially, one static and three flyable prototypes (bort no. 901, 902, 904) were built between 2007 and 2009.^[59] The third one (bort no. 904) was later destroyed when it crashed into a barrier during its taxi runs.^[60]

The first contract for 48 production aircraft was signed at the 2009 MAKS Air show in Moscow. In May 2011, Sukhoi delivered the first Su-35S to Akhtubinsk to conduct state joint tests with the Defence Ministry to prepare for operational service.^[76] The first of two stages of the trials commenced in August 2011. By March 2012, the two prototypes and four production aircraft were conducting flights to test the type's technical characteristics,^[76] which were assessed by the end

of that year to have generally complied with requirements. A batch of six production aircraft was handed over in December 2012.^[77] In February 2013, five of these at the Gromov Flight Research Institute in Zhukovsky started the second stage of the trials, focusing on the Su-35's weapons and combat maneuverability.^[78]

Twelve production Su-35Ss were delivered in December 2013,^[79] followed by another twelve production aircraft in February 2014, ten of which were handed over to the 23rd Fighter Aviation Regiment stationed in the Far East with the remaining two tasked with carrying out the final phase of state joint tests.^[72] The handover marked its official entry into operational service.^[80] Several Su-35Ss were later transferred to Lipetsk to further develop combat tactics and to train service personnel.^[81] The Russia's Su-35Ss are also permanently based at Besovets air base near the Finnish border,^[82] and at Centralnaya Uglovaya air base near Vladivostok.^[83]

The introduction of the Su-35S into the service with the Russian Aerospace Forces is a part of the Russia's state armament programme for 2011–2020 that was formulated following the war with Georgia in 2008 with an aim to significantly increase the number of modern military equipment in the Russian Armed Forces.^[84] The aircraft is delivered alongside the Su-30M2 and Su-30SM and the heavier Su-34 strike aircraft.^[85] The first two are domestic variants of KnAAPO's Su-30MK2 and Irkut's Su-30MKI two-seat export aircraft. According to reports, the simultaneous acquisition of three fighter derivatives of the original Su-27 was to support the two aircraft manufacturers amidst a slump in export orders.^[80] The Su-30M2 serves as a trainer aircraft for the Su-35.^[85]



Sukhoi Su-35S taxiing with drogue parachute after landing during Aviadarts contest, 2019

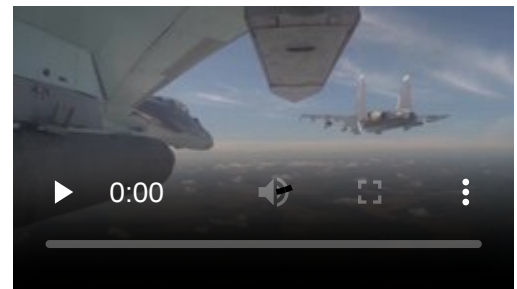
The Su-35S attained full operational capability (FOC) in late 2018.^{[86][87][88]}

In July 2020, the Russian Knights aerobatic team received eight new Su-35Ss as an addition to its previously acquired Su-30SM aircraft.^[89]

The 185th Combat Training and Combat Application Center, also known as an aggressor squadron of the Russian Aerospace Forces, has received three new Su-35S on 9 September 2022.^[90]

Syrian Civil War

In January 2016, Russia made the first combat deployment of the Su-35S when it sent four aircraft to Syria. This occurred following the increased tensions between Russia and Turkey as result of reported incursions by Russian aircraft into the Turkish airspace and the shooting down of a Russian Su-24 bomber by a Turkish F-16 fighter in November 2015.^{[91][92]} Its combat deployment to Syria helped to find and subsequently resolve number of problems, for example with the aircraft avionics.^[93] Su-35s in Syria have been seen carrying unguided bombs, with Russian sources claiming that the Su-35 has carried out strikes against ground targets using guided weapons.^[91]



A Russian Defence Ministry video of the Su-35S

On 20 August 2019, two Russian Aerospace Forces Su-35Ss, operating from the Khmeimim Air Base, intercepted two Turkish Air Force F-16s over the southern Idlib and forced them to leave the Syrian airspace.^[94] Russian Su-35Ss again intercepted several Israeli aircraft over southern Syria on 10 September 2019 and prevented them from carrying out airstrikes.^[95] Another interception allegedly occurred on 19 September 2019, when two Russian Su-35Ss intercepted Israeli aircraft preparing to attack suburbs of Damascus.^[96] On 15 October 2019, a Turkish F-16 fighter was intercepted by Russian Su-35Ss and forced to retreat as it was attempting to bomb the Syrian Democratic Forces's headquarters in Manbij.^[97] On 12 November 2019, Russian Su-35Ss intercepted an Israeli fighter during airstrikes on Damascus.^[98] On 7 December 2019, several Israeli aircraft were intercepted by Russian Su-35Ss and forced to retreat during an attempt to bomb the T-4 Airbase.^[99]



A Su-35S intercepting a US MQ-9 Reaper over Syrian airspace in 2023.

Russo-Ukrainian War

Russian Su-30SM and Su-35S fighters were used for air superiority missions during the war. By September 2022, at least seven air-to-air victories were confirmed over Ukrainian fighters^[100] and one over a Ukrainian Naval Aviation Mil Mi-14.^[101]

On 6 March 2022, a Su-35S pilot from the 4th Centre for Combat Employment and Retraining of Personnel shot down two Ukrainian Mi-8 helicopters as well as an Su-25 attack aircraft in one sortie during the Battle of Mykolaiv.^{[102][103][104][105]}

On 3 April 2022, a Russian Su-35S was shot down by Ukrainian forces, with the pilot ejecting and being captured; the pilot stated that his Su-35S was shot down near Izyum while battling Ukrainian air defenses.^{[106][107]}

On 19 July 2022, the Ukrainian Air Force command claimed that a Su-35 was shot down by Ukrainian air defences near Kakhovka, but at the time there were no photos confirming the loss.^{[108][109]} In early February 2023, photos of the crash site emerged, confirming the crashed plane was an Su-35S.^[110]

On 14 May 2023, a Ukrainian MIM-104 Patriot missile shot down a Su-35 fighter over Bryansk region.^[111] On 22 May 2023, another Ukrainian MIM-104 Patriot missile shot down a Su-35 over Bryansk region.^[112]

On 28 September 2023, a Russian S-300 surface-to-air missile shot down a friendly Su-35 near the Tokmak region. The preliminary report suggests the S-300 missile's fire control radar could not identify friend or foe, leading to a friendly fire incident that downed the Su-35.^{[113][114]}

Ministry of Defense of Ukraine claimed that Ukrainian air defences shot down two Su-35s in February 2024.^[115] According to them this was its "greatest results in downing Russian jets since October 2022" when they claimed to have downed ten Su-34s and one A-50 that month.^[115]

According to the International Institute for Strategic Studies in May 2023, Russia has lost more than five Su-35 fighters since the beginning of the Ukraine war, either to friendly fire, crashed^[116] or Ukrainian forces shooting down^[117] the aircraft using Western-supplied air defences.^{[118][119]}

On 7 June 2025, video emerged of a burning Russian Su-35 wreckage in the Kursk direction, with the Ukrainian Air Force stating it was shot down by Ukrainian forces.^{[120][121]}

China

During the early 1990s, sales arrangements for the Su-27M were discussed with China. In 1995, Sukhoi officials announced a co-production proposal contingent on Beijing's agreement to purchase 120 aircraft.^[122] However, the Russian Foreign Ministry allegedly blocked the sale of the Su-27M and Tupolev Tu-22M bomber over concerns about the arrangements for Chinese production of the Su-27 derivatives.^[123]

In November 2015, China became the Su-35's first export customer when the Russian and Chinese governments signed a contract worth \$2 billion to buy 24 aircraft for the People's Liberation Army Air Force.^{[124][125]} This deal drew international commentaries due to its significance. David Ochmanek of RAND Corporation said the deal "serves both countries", as Russia relied on foreign sales to maintain its military production capacity, while China sought to enhance its military capabilities through the advanced weapon platforms such as the Su-35.^[126] However, the low number of purchases also signified that the deal was symbolic in nature, as the two countries are still competing over influence in Central Asia, according to Moritz Rudolf of Mercator Institute for China Studies.^[126] Ben Moores of IHS believed the Su-35 deal would be the last major order from China, as the Sino-Russian military cooperation saw "very little action or real substance. China doesn't need Russia as much as Russia needs China."^[126]

Chinese officials had reportedly first shown interest in the Su-35 in 2006,^[127] it was not until Zhuhai Airshow 2010 that Alexander Mikheyev of the Rosoboronexport, the Russian state agency responsible for the export and import of defence products, signified that Russia was ready to start talks with China over the Su-35.^{[128][129]} Russian officials publicly confirmed that talks had been going on in 2012, when a protocol agreement on the purchase was signed.^[130] There were subsequent reports of the two countries signing a contract and of imminent deliveries,^{[131][132]} but negotiations would not actually conclude until 2015.

Sales discussions were protracted due to intellectual property rights concerns. China had reverse engineered the Su-27SK and Su-33 to create the J-11B and J-15, respectively,^[133] there were fears of China copying the airframe and offer the copied design on the export market. At one stage, Rosoboronexport demanded that China issue a legally binding guarantee against copying.^[133] Chinese industry was reportedly interested in the AL-41F1S engine and Irbis-E radar.^{[130][134]} According to *The Diplomat*, China held a specific interest in the Su-35's engine, and was already test-flying the J-11D, which reportedly has less range, payload, and maneuverability than the Su-35 but has an active electronically scanned array radar instead of the less advanced PESA radar of the Su-35.^[135] Rosoboronexport insisted on China purchasing a minimum of 48 aircraft to offset risks of copying; after the Kremlin intervened in 2012, the minimum quantity was lowered to 24.^[136] Another problem was China's insistence that the Su-35 include Chinese-made components and avionics. The Kremlin again intervened and conceded to this demand, allowed the deal to proceed; it was viewed as a major concession since the sales of such components are reportedly lucrative.^[130] The contract did not include any technology transfer.^[127]

The Chinese military received the first four aircraft in December 2016.^{[137][138]} Followed by the first delivery, the People's Liberation Army's website opined that with the J-20's commissioning, Russia understood that the Su-35 would "lose its value on the Chinese market in the near future...we hope very much that Su-35 will be the last (combat) aircraft China imports."^[139] China has received a second batch of ten aircraft in 2017,^[140] and another ten in 2018.^[3] The Su-35S entered service with PLAAF in April 2018,^[141] and are based in Guangdong province in southeast China.^[138] In June 2019, Russia offered China a second batch of Su-35s.^[142] According to *Asia Times*, Chinese source remarked further purchase of Su-35 is not feasible as Russian jet offers little value over the Chinese J-16, due to the latter having superior radar, avionics and other equipment.^[143]

On 20 September 2018, the U.S. imposed sanctions on China's Equipment Development Department and its director, Li Shangfu, for engaging in "significant transactions" with Rosoboronexport, specifically naming China's purchase of ten Su-35s in 2017 as well as S-400 surface-to-air missile system-related equipment in 2018.^[144]

Iran

Following the Russian purchase of Iranian suicide drones during the invasion in Ukraine, the Islamic Republic of Iran Air Force (IRIAF) is reportedly considering the procurement of the Sukhoi Su-35 in exchange for further delivery of ammunition, drones and ballistic missiles to the Russian Armed Forces, according to US officials. It was also reported that training of Iranian pilots has begun in Spring 2022 and that the aircraft may be delivered in 2023. The deal may include 24 Su-35s originally built for Egypt, but no official reports of delivery have emerged so far.^{[145][146]} The deal was confirmed by Iranian state-owned media Tasnim News Agency on 28 December 2022 and on 15 January 2023 by an Iranian MP who said that supplies will begin in the spring and also said that Iran has ordered from Russia air defense and missile systems as well as helicopters.^[147]

However, in July 2023, the Iranian Defense Minister, Mohammad-Reza Gharaei Ashtiani, made vague statements alluding that Iran is reconsidering the acquisition of the Su-35 due to increasing investment towards indigenous fighter building capabilities, leading to some sources to declare that the deal has collapsed.^[148] Despite this, the Iranian Air Force has received deliveries of the Yak-130 advanced jet trainer, which are used to train aspiring cadets before they graduate to actual fighter aircraft, such as the

Su-35. The trainers have received Islamic Republic of Iran Air Force (IRIAF) livery and serial numbers. On 28 November 2023, Iran's deputy defense minister announced that Iran had finalized a deal for the purchase of Su-35 aircraft from Russia, although further details on the quantity or timeline were not shared.^[149] An order for 24 Su-35s was reported in April 2024,^[150] but these reports were denied by Iran's SNN media agency.^[151] In January 2025, Ali Shadmani, a senior commander in the Islamic Revolutionary Guards Corps, said that Su-35s had been purchased by Iran, without precisising how many were procured and whether deliveries had started.^[152]

An Iranian lawmaker said on 23 September 2025 that Su-35s are "on the way".^[153]

Potential operators

United Arab Emirates

In the mid-1990s, the United Arab Emirates evaluated the Su-27M,^[154] but later acquired the Mirage 2000 due to the country's close relationship with France.^[27] In February 2017, the country was to sign a preliminary agreement for the Su-35 purchase and also signed an agreement with Rostec, Russia's state-owned corporation responsible for the development of advanced industrial products, to develop a fifth-generation aircraft based on the MiG-29.^[155]

India

India has been reluctant to order the Sukhoi/HAL FGFA due to high cost, and it has been reported that India and Russia are studying an upgrade to the Su-35 with stealth technology (similar to the F-15 Silent Eagle) as a more affordable alternative to the FGFA (Su-57).^[156] The aircraft is competing with 7 other aircraft in a fresh tender which is referred as MMRCA 2.0 in the Indian media, for the procurement of 114 multi-role combat aircraft.^[157]

Algeria

According to *Kommersant*, the Algerian military had requested a Su-35 for testing in February 2016; it was reported the country was satisfied with the fighter's flight characteristics and so Moscow is waiting for a formal application.^[158] On 27 December 2019, Algeria has reportedly signed a contract for 14 aircraft as part of large military deal that also includes purchase of Su-34 and Su-57 fighters.^[159] However, neither Russian nor Algerian governments have confirmed that such deal exists.

Turkey

In July 2019, Russia offered the Su-35 to Turkey after it was removed from the F-35 programme by United States due to the purchase of the S-400 missile system.^[160] CEO of Rostec Sergey Chemezov, said, "If our Turkish colleagues express a desire, we are ready to work out the deliveries of Su-35 fighter jets".^[161] Previously in May 2019, Sergey Chemezov had said that Russia was ready to cooperate with Turkey on the export and local production of the Su-57.^[162]

Vietnam

The New York Times alleges that Vietnam has been quietly pursuing Su-35 procurement, featuring sanctions evasion and indirect payments.^[163]

Others

Following the deployment to Syria of several new Russian military systems, various countries had reportedly expressed interest in the Su-35. These countries included Algeria, Egypt, and Vietnam.^{[158][164][165]} Other countries that had also expressed interest in the aircraft include Kazakhstan,^[166] North Korea,^[167] and Pakistan although a Russian official denied that the country was in talks with the latter about the Su-35. Sudan has reportedly also expressed an interest in acquiring of the Su-35 fighters during the Sudanese president Omar Hassan al-Bashir's visitation of Moscow in November 2017.^[168]

Failed bids

Brazil

In the mid-1990s, Brazilian and Russian authorities conducted talks on the possible acquisition of the Su-27M.^[169] In 2001, the Brazilian government launched the F-X tender, the objective of which was to procure at least 12 aircraft to replace the Brazilian Air Force's ageing aircraft, primarily the Mirage IIIs.^{[170][171]} Since the Brazilian government was also looking to develop the country's aerospace and defence industries, Sukhoi partnered with the Brazilian defence contractor Avibras during the tender. The two companies submitted the Su-27M to the US\$700-million tender, and included an offset agreement wherein the Brazilian industry would have participated in the manufacturing of certain aircraft equipment.^[172] The tender was suspended in 2003 because of domestic political issues and then scrapped in 2005, pending the availability of new fighters.^[170] The Su-27M was preferred over the next favourite, the Mirage 2000BR.^[171] had the aircraft been purchased, it would have been the first heavy fighter delivered to Latin America.^[169]

With the tender relaunched in 2007 as the F-X2 competition, the Brazilian Defence Ministry looked to purchase at least 36 aircraft – with a potential for 84 additional aircraft – to replace the country's A-1Ms, F-5BRs and Mirage IIIs. Among the participants were the F/A-18E/F Super Hornet, F-16BR, JAS Gripen NG, Dassault Rafale, Eurofighter Typhoon and the modernized Su-35. Although the Brazilian government eliminated the Su-35 in 2008,^[173] Rosoboronexport subsequently offered to sell the country 120 aircraft with full technology transfer,^[174] as well as participation in the PAK FA programme.^[175] In December 2013, the Gripen NG light fighter was selected because of its low cost and the transfer of technology to the Brazilian industry.^[176]

Egypt

In March 2019, it was reported that Egypt would procure "over two dozen" Su-35s from Russia in a deal valued about \$US2 billion, finalized in late 2018. Deliveries were expected to begin as early as 2020 or 2021.^{[177][178]} Deputy head of the Rosoboronexport Sergei Kornev denied the signing of any contract to supply Su-35s to Egypt.^[179] U.S. Secretary of State Mike Pompeo warned Egypt against buying Su-35s, saying "if those systems were to be purchased, the CAATSA statute would require sanctions on the regime."^[180]

On 19 May 2020, Russia began Su-35 production for Egypt,^[181] and first five production aircraft took off from the KnAAZ plant on 28 July 2020.^[182] By June 2021, in total 17 aircraft were produced for Egypt.^{[4][183]} Egyptian newspaper Arabo reported on 5 January 2022, Egypt, Algeria and Indonesia announced that they would cancel their Su-35 orders with Russia. In March 2022, an agreement with the

US to sell variants of F-15Es to the Egyptian Air Force was announced which confirms that Egypt either abandoned or cancelled the Su-35 deal and the planes originally produced for Egypt will be delivered to Iran instead.^{[184][185][186]} A contract needs to be finalized after price and delivery date are determined.^[187]

Indonesia

In 2014, Russia offered the Su-35 for Indonesia's Northrop F-5E replacement competition.^[188] In 2015, the Su-35 was selected based on the Indonesian Air Force's familiarity with the Su-27SK and Su-30MK2; the other competitors were the Eurofighter Typhoon, Dassault Rafale, F-16 and Saab JAS 39 Gripen.^{[189][190]} A contract for 11 aircraft was signed in February 2018;^[191] half of the US\$1.14 billion price was to be being paid for in commodities.^[191] The first two aircraft were expected to be delivered in October.^[191]

As of May 2024, no aircraft have been delivered. In March 2020, Bloomberg reported the purchase was cancelled due to American pressure;^[192] Indonesia^[193] and Russia denied the cancellation.^[194] In December 2021, Chief of Staff of the Indonesian Air Force Air Chief Marshal Fadjar Prasetyo said that the purchase would be cancelled for budgetary reasons.^[195] In June 2023, the Indonesian Ministry of Defense reported that the purchase was hampered by the threat of American sanctions through CAATSA and OFAC.^[196] In May 2024, Jose Tavares, the Indonesian Ambassador to Russia, stated that the contract remained in force.^[197]

Others

In 1996, Russia submitted the Su-27M and Su-37 for South Korea's F-X programme, which sought a 40-aircraft replacement for the Republic of Korea Air Force's F-4D/Es, RF-4Cs and F-5E/Fs. The two Russian designs competed against the Dassault Rafale, Eurofighter Typhoon and F-15K Slam Eagle.^[198] Sukhoi proposed a design which featured a phased-grid radar and thrust-vectoring engines, and offered full technology transfer as well as final assembly in South Korea. The US\$5 billion contract would have been partially financed through a debt-reduction deal on money Russia owed to South Korea.^{[199][200]} However, the Su-27M was eliminated early in the competition, which was won by the F-15K.^[201]

A country that had been reported to be a likely early export customer for the modernized Su-35 was Venezuela. The Venezuelan government of Hugo Chávez in July 2006 placed an order for 24 Su-30MK2s to replace its fleet of F-16s that were subjected to a US arms embargo.^[202] The aircraft were delivered to the Venezuelan Air Force from 2006 to 2008. The country was expected to follow up with a second order for the same type, or make a purchase of the Su-35.^[203] Despite subsequent reports that the Venezuelan government were interested in the aircraft and had placed an order for the Su-35,^[204]

Libya was also expected to be an early export customer for 12 to 15 Su-35s along with other Russian weapons; however, the civil war in Libya and the resulting military intervention cancelled such plans.^[205] Russia has also offered the modernized Su-35 to India, Malaysia, and Greece;^[206] no firm contracts have materialised, with the first two countries having been occupied with other fighter projects and unlikely to procure the modernized Su-35.^[39]

Variants

Su-27M/Su-35

Single-seat fighter design with a factory code of T-10M (*Modernizirovany*, "Upgraded"). The first two prototypes had a new forward fuselage, canards and updated fly-by-wire flight-control systems. Like three of KNAAPO's nine flying pre-production aircraft (T10M-5, T10M-6 and T10M-7), they were converted from Su-27 airframes.^{[207][208]} The third aircraft (T10M-3) was the first of seven pre-production aircraft to have the taller vertical tails, two-wheel nose undercarriage and in-flight refuel capability.^[20] The Su-27M was powered by AL-31FM turbofan engines.^[18] Two prototypes, nine pre-production and three production aircraft were constructed by 1995,^[1] two static-test aircraft was also constructed (T10M-0 and T10M-4).^[209] The aircraft did not enter mass production.

Su-37

Technology demonstrator, converted from the eleventh developmental Su-27M (T10M-11). The Su-37 featured a digital fly-by-wire flight-control system, a glass cockpit, the N011M radar, and AL-31FP engines with thrust-vectoring nozzles.^[210] The aircraft was later fitted with standard-production AL-31F engines, and had its flight-control system and cockpit systems revised.^[211]



Sukhoi Su-37 at Farnborough 1996 airshow

Su-35UB

Two-seat trainer designed and built by KnaAPO. The single aircraft (T-10UBM-1) featured the canards and taller vertical tails of the Su-27M and a forward fuselage similar to the Su-30MKK. The Su-35UB also shared the avionics suite of the Su-30MKK, although it had a different fly-by-wire flight-control system to accommodate the canards.^[212] The aircraft was powered by AL-31FP engines with thrust-vectoring nozzles.^[213] Although a training aircraft, the Su-35UB was designed to be fully combat-capable.^[212]



The sole two-seat Su-35UB in 2001

Su-35BM

Single-seat fighter that is a major redesign of the original Su-27. The type features significant modifications to the airframe, including the removal of canards and dorsal air brake as found on the Su-27M. It features the updated N035 Irbis-E radar and a redesigned cockpit. The aircraft is powered by thrust-vectoring AL-41F1S turbofan engines that are capable of supercruise. Also known as T-10BM (*Bolshaya Modernizatsiya*, "Major Modernization"), Su-35BM is not the actual designation used by Sukhoi, who markets the aircraft as "Su-35".^[214]

Su-35S

Su-35S is the production designator for the T-10BM design for the Russian Aerospace Forces. According to *Aviation Week & Space Technology*, "S" stands for *Stroyevoy* ("Combatant").^[134] NATO reporting name Flanker-M.^[215] The Flanker M is the latest version of the Sukhoi Su-35, known internally as the Su-35BM (*bolshaya modernizatsiya* – big modernization), and is an advanced capability multirole air superiority fighter developed from the Su-27. The aircraft has high maneuverability (+9g) with a high angle

of attack, and is equipped with high-capability weapon systems that contribute to the new aircraft's exceptional dogfighting capability.^[215] The Russian Air Force ordered 48 Sukhoi Su-35S fighters in August 2009 with deliveries scheduled to run until 2015. Sukhoi began producing the components required for assembly of the aircraft in November 2009. An information management system integrated with onboard subsystems and a new phased array radar system with long-range aerial target detection have been installed in the aircraft. The first Su-35S aircraft was handed over to the 929th State Flight Test Centre (GLITS) for flight tests in August 2011. The Russian Ministry of Defence received six Su-35S production aircraft from Sukhoi in December 2012. In 2015, China ordered 24 Su-35S aircraft for China's People's Liberation Army Air Force.^{[215][216]}



A combat-ready Su-35S equipped with pairs of R-73, R-27 and R-77 air-to-air missiles in an exercise.

Su-35SM

Su-35SM is a upgrade to the Su-35S, designed for the Russian Aerospace Forces. This variant aims to enhance air superiority and electronic warfare capabilities. Notable upgrades include an AESA radar, similar to that used in the Su-57, which will replace the Irbis-E radar, boosting detection range and resistance to countermeasures. The aircraft will also feature the "Himalaya" electronic warfare suite, offering advanced protection against radar-guided threats. New secure communications and data-link systems are included for better network-centric operations, along with a laser-based DIRCM system to counter infrared missiles. The airframe may use materials to lower radar visibility and be optimized for next-generation munitions like the R-37M and R-77M missiles.^[217]

Operators

Algeria

- Algerian Air Force - 24 Su-35s on order with first units received in March 2025.^{[218][219]}
 - Oum el Bouaghi Air Base, Oum el Bouaghi Province

People's Republic of China

- People's Liberation Army Air Force – 24 aircraft in inventory.^[3]
 - 6th Aviation Brigade – Suixi air base, Guangdong^[138]

Iran

- Islamic Republic of Iran Air Force - 50 Su-35s on order.^{[220][221]}

Russia



Su-35 Operators

- Russian Aerospace Forces – 110 aircraft in inventory as of December 2022.^{[222][2][223][224][225]} New deliveries conducted in June, July, September and October 2023 for a probable total quantity of 8 delivered aircraft.^{[226][227][228][229]} The final 2023 delivery was conducted in November^[230] and the first one for 2024 took place in April.^[231] The second delivery was conducted in May^[232] and the third one in November.^[233] For 2025, the first delivery was conducted in March,^[234] the second in May,^[235] the third in June,^[236] the fourth in August,^[237] the fifth in September,^[238] and the sixth in November.^[239] The third order for 30 aircraft finalized in August 2020 is to increase the total number to 128.^{[68][240][241]}



Su-35S of the "Russian Knights" in Langkawi for LIMA 2023

- 23rd Fighter Aviation Regiment – Dzyomgi Airport, Khabarovsk Krai^[72]
- 22nd Guards Fighter Aviation Regiment – Centralnaya Uglovaya Air Base, Primorsky Krai^[83]
- 159th Fighter Aviation Regiment – Besovets Air Base, Republic of Karelia
- 790th Fighter Aviation Regiment – Borisovsky Khotilovo Air Base, Tver Oblast^{[242][243]}
- 4th Centre for Combat Employment and Retraining of Personnel – Lipetsk Air Base, Lipetsk Oblast
- 929th State Flight-Test Centre – Vladimirovka Air Base, Astrakhan Oblast^[244]
- Khmeimim Air Base, Latakia, Syria^[245]

Notable accidents

The fourth Su-35BM prototype was destroyed in a high-speed taxi run on 26 April 2009 due to brake failure. The aircraft crashed into the barrier at the end of the runway and was destroyed. The pilot ejected and sustained burn injuries.^[60]

On 31 July 2021, a Su-35S fighter crashed after suffering an engine failure during a routine training mission in the Khabarovsk Territory in Russia, according to statement from the Russian Defence Ministry.^[246] According to a preliminary investigation reported by the Russian newspaper Top War, the Su-35 suffered technical malfunctions in the environment control systems that indicate heating and cooling functions.

On 28 March 2024, a Su-35 crashed into the sea off Sevastopol. The pilot is reported to have safely ejected.^[247]

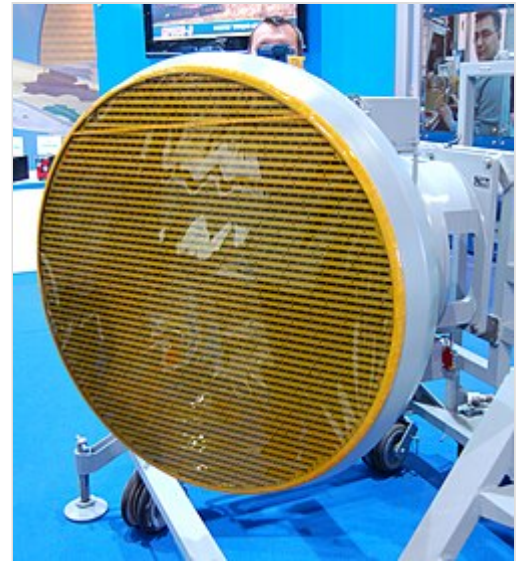
Specifications (Su-35S)

Data from KnAAPO,^{[45][248]} Jane's All The World's Aircraft 2013^[249]

General characteristics

- **Crew:** 1
- **Length:** 21.9 m (71 ft 10 in)

- **Wingspan:** 15.3 m (50 ft 2 in)
- **Height:** 5.9 m (19 ft 4 in)
- **Wing area:** 62 m² (670 sq ft)
- **Airfoil:** 5%
- **Empty weight:** 19,000 kg (41,888 lb) ^[250]
- **Gross weight:** 25,300 kg (55,777 lb) with 50% internal fuel
- **Max takeoff weight:** 34,500 kg (76,059 lb)
- **Fuel capacity:** 11,500 kg (25,400 lb) internal
- **Powerplant:** 2 × Saturn AL-41F1S afterburning turbofan engines, 86.3 kN (19,400 lbf) thrust each dry, 137.3 kN (30,900 lbf) with afterburner, 142.2 kN (32,000 lbf) in emergency power



Irbis-E radar for the modernized Su-35 at MAKS Airshow 2009

Performance

- **Maximum speed:** 2,400 km/h (1,490 mph, 1,290 kn) at altitude
- **Maximum speed:** Mach 2.25
 - 1,400 km/h (870 mph; 760 kn), Mach 1.13 at sea level
- **Cruise speed:** 1,170 km/h (730 mph, 630 kn) / Mach 1.1+ supercruise at medium altitude^[251]
- **Range:** 3,600 km (2,200 mi, 1,900 nmi) at altitude
 - 1,580 km (982 mi) at sea level
- **Combat range:** 1,600 km (990 mi, 860 nmi) approx^[252]
- **Ferry range:** 4,500 km (2,800 mi, 2,400 nmi) with 2 external fuel tanks
- **Service ceiling:** 18,000 m (59,000 ft)
- **g limits:** +9
- **Rate of climb:** 280 m/s (55,000 ft/min) +
- **Wing loading:** 408 kg/m² (84 lb/sq ft) With 50% fuel
 - 500.8 kg/m² (102.6 lb/sq ft) with full internal fuel
- **Thrust/weight:** 1.13 with 50% fuel
 - 0.92 with full internal fuel



GSh-30-1 cannon in starboard wing root, Su-35, Paris Air Show 2013

Armament

- **Guns:** 1 × internal 30 mm Gryazev-Shipunov GSh-30-1 autocannon with 150 rounds
- **Hardpoints:** 12 hardpoints, consisting of 2 wingtip rails, and 10 wing and fuselage stations with a capacity of 8,000 kg (17,630 lb) of ordnance, with provisions to carry combinations of:
 - **Rockets:**
 - S-25 air-ground rocket
 - **Missiles:**

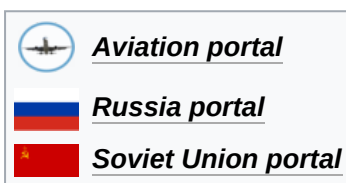
- Air-to-air missiles:
 - 8 × R-27ER/ET/EA
 - 6 × R-73E/M
 - 12 × R-77M/P/T/-1
 - 6 × R-74
 - 2 × R-37^[253]
- Air-to-surface missiles:
 - 6 × Kh-29L/TE
 - 3 × 3M-14AE
- Anti-ship missiles:
 - 3 × 3M-54AE1
 - 6 × Kh-31A/AD
 - Kh-35U
 - 5 × Kh-59MK
 - 1 × P-800 Oniks
- Anti-radiation missiles:
 - 6 × Kh-31P/PD
 - 5 × Kh-58USHE
- **Bombs**:
 - 8 × KAB-500KR TV-guided bombs
 - 8 × KAB-500L laser-guided bombs
 - 8 × KAB-500OD guided bombs
 - 8 × KAB-500S-E satellite-guided bombs
 - 3 × KAB-1500KR TV-guided bombs
 - 3 × KAB-1500L laser-guided bombs

Avionics

- Irbis-E passive electronically scanned array radar
- OLS-35 infra-red search and track system
- L175M Khibiny-M electronic countermeasures system^[47]

Notable appearances in media

See also



Related development

- [Sukhoi Su-27](#)
- [Sukhoi Su-30](#)
- [Sukhoi Su-33](#)
- [Sukhoi Su-34](#)
- [Sukhoi Su-37](#)
- [Sukhoi Su-57](#)^[N 2]

Aircraft of comparable role, configuration, and era

- [Boeing F/A-18E/F Super Hornet](#)
- [McDonnell Douglas F-15E Strike Eagle](#)
- [Boeing F-15EX Eagle II](#)
- [Dassault Rafale](#)
- [Mikoyan MiG-35](#)
- [Eurofighter Typhoon](#)
- [Shenyang J-11D](#)
- [Shenyang J-16](#)

Related lists

- [List of fighter aircraft](#)
- [List of military aircraft of the Soviet Union and the CIS](#)

Footnotes

1. "While the canard layout brought advantages in terms of improved maneuverability, it also added structural weight to the airframe. A conventional airframe coupled with thrust vector control, the source said, could now provide the same capability."^[40]
2. Quote: "Another objective of Sukhoi's programme for development of the Su-35BM is gradual testing and implementation of design features prepared for PAK FA (Perspektivnyi Aviatsionnyi Kompleks Frontovoi Aviatsii or Future Aviation Complex for Tactical Aviation)."^[7]

References

1. Butowski 2004, p. 38.
2. "Russian Defense Ministry receives last three Su-35S fighters under 5-year contract" (<http://tass.com/defense/1227585>). *TASS*. 25 November 2020. Retrieved 26 June 2021.
3. "Russia Completes Delivery of 24 Su-35 Fighter Jets to China" (<https://thediplomat.com/2019/04/russia-completes-delivery-of-24-su-35-fighter-jets-to-china/>). *The Diplomat*. 17 April 2019. Retrieved 23 February 2020. China: 24
4. "Egyptian Air Force will receive Su-35 fighter jets" (<https://web.archive.org/web/20230802074838/https://airrecognition.com/index.php/news/defense-aviation-news/2021/june-2021/7400-egyptian-air-force-already-received-17-su-35-fighter-jets.html>). *airrecognition.com*. 21 June 2021. Archived from the original (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2021/june-2021/7400-egyptian-air-force-already-received-17-su-35-fighter-jets.html>) on 2 August 2023. Retrieved 26 June 2021.
5. "Партия новых самолетов Су-35с поступила в Липецкий авиационный ВКС : Министерство обороны Российской Федерации" (https://function.mil.ru/news_page/country/more.htm?id=12398526@egNews). Retrieved 17 March 2023.

6. Jennings, Gareth (19 June 2013). "Paris Air Show 2013: Le Bourget debut for Su-35". *IHS Jane's Defence Weekly*. **50** (29). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>). "Also referred to as the Su-35S 'Super Flanker' (and sometimes the Su-35C), the Sukhoi was originally conceived purely for export sales, with China and Venezuela expressing interest."
7. Butowski, Piotr (22 June 2004). "Halfway to PAK FA: the latest addition to the Su-27 family seeks to bridge the gap to the fifth generation combat aircraft" (<https://web.archive.org/web/20150924122311/http://www.highbeam.com/doc/1G1-126558892.html>). *Interavia Business & Technology*. Aerospace Media Publishing. OCLC 71365085 (<https://search.worldcat.org/oclc/71365085>). Archived from the original (<http://www.highbeam.com/doc/1G1-126558892.html>) on 24 September 2015. Retrieved 19 April 2013.
8. Butowski, Piotr (1 November 1999). "Dominance by design: the reign of Russia's 'Flankers' – PART ONE". *Jane's Intelligence Review*. **11** (11). Coulsdon, UK: Jane's Information Group. ISSN 1350-6226 (<https://search.worldcat.org/issn/1350-6226>).
9. Gordon 2007, p. 69.
10. Andrews 2003, p. 39.
11. Gordon 2007, pp. 58, 122.
12. Fink 1993, p. 45.
13. Gordon 2007, pp. 122–123.
14. Williams 2002, p. 119.
15. Gordon 2007, pp. 123, 127.
16. Gordon 2007, p. 124.
17. Gordon 2007, p. 69.
18. Gordon 2007, p. 123.
19. Gordon 2007, pp. 126–127.
20. Gordon 2007, p. 128.
21. "Cy-35" (<http://www.airwar.ru/enc/fighter/su35.html>). *airwar.ru*. Retrieved 23 February 2020.
22. Gordon 2007, p. 366.
23. Gordon 2007, pp. 129–131.
24. Fink 1993, p. 44.
25. Gordon 2007, pp. 134–135.
26. *Flight International* 2001, p. 20.
27. Andrews 2003, pp. 58.
28. Gordon 2007, pp. 142, 144.
29. Velovich 1996, p. 16.
30. Novichkov 1996, p. 52.
31. Gordon 2007, pp. 172–173.
32. Barrie 1997, p. 8.
33. Andrews 2003, p. 47.
34. Gordon 2007, p. 173.
35. Fiszler 2004, pp. 52–53.
36. Fiszler 2004, p. 53.
37. Butowski 2007, pp. 34–35.
38. Butowski, Piotr (10 May 2021). "Sukhoi Su-35 The Ultimate Flanker" (<https://www.key.aero/article/sukhoi-su-35-ultimate-flanker>). *Key Publishing*.

39. "Russia's Su-35 Super-Flanker: Mystery Fighter No More" (<https://web.archive.org/web/20130501070942/http://www.defenseindustrydaily.com/Russias-SU-35-Mystery-Fighter-No-More-04969/>). *Defense Industry Daily*. 27 March 2013. Archived from the original (<http://www.defenseindustrydaily.com/russias-su-35-mystery-fighter-no-more-04969/>) on 1 May 2013. Retrieved 20 April 2013.
40. Barrie 2003, p. 39:
41. Butowski 2004, p. 39.
42. Karnozov, Vladimir (4 September 2007). "Sukhoi unveils 'supercruising' Su-35-1 multi-role fighter" (<https://web.archive.org/web/20170808021053/https://www.flightglobal.com/news/articles/sukhoi-unveils-su-35-l-216386/>). *Flightglobal*. Archived from the original (<https://www.flightglobal.com/news/articles/sukhoi-unveils-su-35-l-216386/>) on 8 August 2017. Retrieved 22 July 2017.
43. Kopp 2010, p. 9.
44. "Su-35" (<https://web.archive.org/web/20130502190948/http://sukhoi.org/eng/planes/military/Su-35/>). Sukhoi. Archived from the original (<http://www.sukhoi.org/eng/planes/military/Su-35/>) on 2 May 2013. Retrieved 25 April 2013.
45. "Su-35: Multifunctional Super-Maneuverable Fighter" (https://web.archive.org/web/20130921083835/http://www.knaapo.ru/media/eng/about/production/military/su-35/su-35_buklet_eng.pdf) (PDF). KnAAPO. Archived from the original (http://www.knaapo.ru/media/eng/about/production/military/su-35/su-35_buklet_eng.pdf) (PDF) on 21 September 2013. Retrieved 27 March 2013.
46. Gordon 2007, p. 175.
47. "Sukhoi begins flight testing of Su-35S variant" (https://web.archive.org/web/20170830022756/http://www.defencweb.co.za/index.php?option=com_content&view=article&id=15220). *Defencweb.co.za*. 6 May 2011. Archived from the original (http://www.defencweb.co.za/index.php?option=com_content&view=article&id=15220) on 30 August 2017. Retrieved 30 August 2017.
48. Sweetman 2004, p. 26.
49. Alex Lockie (16 September 2016). "Here's who'd win in a dogfight between Russia's and the US's top fighter jets" (<https://www.businessinsider.com/su-35-vs-f-22-2016-9#stealth-6>). *Business Insider*. Retrieved 27 January 2024.
50. Butowski 2007, p. 36.
51. Gordon 2007, pp. 175–176.
52. Bill, Sweetman (24 June 2013). "Sukhoi Test Pilot Explains 'Supermaneuverability' " (https://aviationweek.com/Article.aspx?id=/article-xml/AW_06_24_2013_p33-589854.xml&p=1). *Aviation Week & Space Technology*. Archived (https://web.archive.org/web/20140304215955/http://www.aviationweek.com/Article.aspx?id=%2Farticle-xml%2FAW_06_24_2013_p33-589854.xml&p=1) from the original on 4 March 2014. Retrieved 24 February 2020.
53. Butowski 2004, pp. 39–40.
54. Butowski *Air International* October 2019, p. 37.
55. Sweetman 2013, p. 44.
56. Fomin 2008, p. 28.
57. Usov, Pavel (16 February 2007). На КНААПО поступят двигатели для Су-35 (<https://web.archive.org/web/20130823234739/http://www.kommersant.ru/doc/743121>). *Kommersant* (in Russian). Archived from the original (<http://www.kommersant.ru/doc/743121>) on 23 August 2013. Retrieved 20 April 2013.
58. Начались испытания новой "сушки" (<https://web.archive.org/web/20131029195851/http://www.kommersant.ru/doc/837808>). *Kommersant* (in Russian). 19 December 2007. Archived from the original (<http://www.kommersant.ru/doc/837808>) on 29 October 2013. Retrieved 20 April 2013.

59. Начались испытания второго летного образца новейшего Су-35 (<https://web.archive.org/web/20130824055911/http://www.sukhoi.org/news/smi/arch/index.php?id=1963>) (Press release) (in Russian). Sukhoi. 3 October 2008. Archived from the original (<http://www.sukhoi.org/news/smi/arch/index.php?id=1963>) on 24 August 2013.
60. Nechayev, Gennady (27 April 2009). Полностью разрушился и сгорел (<https://web.archive.org/web/20090430093816/https://vz.ru/society/2009/4/27/280531.html>). *Vzglyad* (in Russian). Archived from the original (<https://vz.ru/society/2009/4/27/280531.html>) on 30 April 2009. Retrieved 27 July 2017.
61. Doronin, Nina (27 April 2009). В ходе испытаний на заводском аэродроме сгорел истребитель Су-35 (<https://web.archive.org/web/20090430102444/http://www.rg.ru/2009/04/27/reg-dvostok/istrebitel-anons.html>). *Rossiyskaya Gazeta* (in Russian). Archived from the original (<http://www.rg.ru/2009/04/27/reg-dvostok/istrebitel-anons.html>) on 30 April 2009. Retrieved 20 April 2013.
62. "В России из-за отказа тормозной системы разбился новейший истребитель Су-35" (<http://news.bigmir.net/world/131925/>). 25 September 2011. Archived (<https://web.archive.org/web/20190410200300/http://news.bigmir.net/world/131925/>) from the original on 10 April 2019. Retrieved 10 April 2019.
63. Истребитель Су-35 разбился в Комсомольске-на-Амуре (<http://www.aviaport.ru/digest/2009/04/27/171683.html>) Archived (<https://web.archive.org/web/20181219182149/https://www.aviaport.ru/digest/2009/04/27/171683.html>) 19 December 2018 at the *Wayback Machine* // [aviaport.ru](http://www.aviaport.ru), 2009
64. Заявление пресс-службы компании «Сухой» в связи с аварией третьего лётного образца истребителя Су-35 (<http://www.aviaport.ru/digest/2009/04/27/171760.html>) Archived (<https://web.archive.org/web/20181219182206/https://www.aviaport.ru/digest/2009/04/27/171760.html>) 19 December 2018 at the *Wayback Machine* // [aviaport.ru](http://www.aviaport.ru), 2009
65. Butowski 2004, p. 40.
66. Gordon 2007, p. 111.
67. Su-35 Begins Global Journey. *Flight Daily News*. Sutton, UK. Reed Business Information. 20 June 2013. pp. 16–17. The head of Sukhoi's parent company, United Aircraft, is forecasting 200 sales of the type, split 50:50 between domestic and export.
68. Pyadushkin, Maxim (12 January 2016). "Russia Places New Order For 50 Su-35S Fighters" (<http://aviationweek.com/defense/russia-places-new-order-50-su-35s-fighters?>). *Aerospace Daily & Defense Report*. Aviation Week. Archived (<https://web.archive.org/web/20170811042649/http://aviationweek.com/defense/russia-places-new-order-50-su-35s-fighters>) from the original on 11 August 2017. Retrieved 29 July 2017.
69. "Sukhoi launches production of Su-35 for Russia" (<https://web.archive.org/web/20110217145255/http://www.defencetalk.com/sukhoi-launches-production-of-su-35-for-russia-23114/>). *Defence Talk*. 24 November 2009. Archived from the original (<http://www.defencetalk.com/sukhoi-launches-production-of-su-35-for-russia-23114/>) on 17 February 2011. Retrieved 27 March 2013.
70. "Sukhoi completed general units' assembly of the first production Su-35S" (<https://web.archive.org/web/20130824060950/http://www.sukhoi.org/eng/news/company/?id=3668>) (Press release). Sukhoi. 11 October 2010. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=3668>) on 24 August 2013. Retrieved 21 April 2013.
71. "Sukhoi is completing preliminary tests of the Su-35" (<https://web.archive.org/web/20101010011358/http://www.sukhoi.org/eng/news/company/?id=3470>) (Press release). Sukhoi. 20 July 2010. Archived from the original (<http://www.sukhoi.org/eng/news/company/?id=3470>) on 10 October 2010. Retrieved 21 April 2013.
72. Fomin 2014, p. 37.

73. "Комсомольские авиастроители выпустили сотый Су-35" (<https://web.archive.org/web/20230927072435/https://uacrussia.ru/ru/press-center/news/komsomolskie-aviastroiteli-vypustili-sotyy-su-35>) (Press release). United Aircraft Corporation. 17 December 2018. Archived from the original (<https://uacrussia.ru/ru/press-center/news/komsomolskie-aviastroiteli-vypustili-sotyy-su-35>) on 27 September 2023. Retrieved 23 February 2020.
74. Gordon 2007, pp. 365–366.
75. Gordon 2007, pp. 137, 141.
76. Fomin 2012, p. 16.
77. "Sukhoi Delivered 6 Su-35S to the Ministry of Defense" (<https://web.archive.org/web/20130824024622/http://sukhoi.org/eng/news/company/?id=5037.html>) (Press release). Sukhoi. 28 December 2012. Archived from the original (<http://sukhoi.org/eng/news/company/?id=5037.html>) on 24 August 2013. Retrieved 26 March 2013.
78. Mikhailov, Alexey (20 February 2013). "BBC начали отработку сверхманевренного ближнего боя на Су-35С" (<https://web.archive.org/web/20130220114434/http://izvestia.ru/news/545232>). *Izvestia* (in Russian). Archived from the original (<http://izvestia.ru/news/545232>) on 20 February 2013. Retrieved 22 April 2013.
79. "TASS: Russia – Sukhoi supplies 12 Su-35 jet fighters to Russian Air Force in 2013" (<http://en.itar-tass.com/russia/712932>). TASS. Archived (<https://web.archive.org/web/20131229043147/http://en.itar-tass.com/russia/712932>) from the original on 29 December 2013. Retrieved 1 April 2015.
80. Newdick, Thomas (21 February 2014). "Russia's New Air Force Is a Mystery" (<https://web.archive.org/web/20140905035938/https://medium.com/war-is-boring/russias-new-air-force-is-a-mystery-af28abb0be7d>). *War is Boring*. Archived from the original (<https://medium.com/war-is-boring/russias-new-air-force-is-a-mystery-af28abb0be7d>) on 5 September 2014. Retrieved 29 July 2017.
81. Fomin 2014, p. 36.
82. "New Generation Fighter Aircrafts [sic] Su-35 Were Added to Karelian Air Regiment" (https://web.archive.org/web/20170906153647/http://www.gov.karelia.ru/News/2016/12/1206_15_e.html) (Press release). Government of the Republic of Karelia. Archived from the original (http://www.gov.karelia.ru/News/2016/12/1206_15_e.html) on 6 September 2017. Retrieved 6 September 2017.
83. "Pilots of the Eastern MD performed aerobatics over the Bay of Ajax in Vladivostok" (https://web.archive.org/web/20170906154537/http://eng.mil.ru/en/news_page/country/more.htm?id=12140932@egNews) (Press release). Russian Ministry of Defence. 5 September 2017. Archived from the original (http://eng.mil.ru/en/news_page/country/more.htm?id=12140932@egNews) on 6 September 2017. Retrieved 6 September 2017.
84. Litovkin, Dmitry (20 January 2009). Армия станет "триллионером" (<https://web.archive.org/web/20170808193413/https://iz.ru/news/344614>). *Izvestia* (in Russian). Archived from the original (<https://iz.ru/news/344614>) on 8 August 2017. Retrieved 8 August 2017.
85. Pyadushkin 2010, p. 28.
86. "Russia's advanced Sukhoi Su-35S fighter put into operation" (<https://web.archive.org/web/20170923051701/http://tass.com/defense/966319>). TASS. 19 September 2017. Archived from the original (<http://tass.com/defense/966319>) on 23 September 2017. Retrieved 23 September 2017.
87. "Государственные испытания истребителя Су-35С завершатся к концу года" (<http://www.armstrade.org/includes/periodics/news/2018/0404/115546147/detail.shtml>). *Arms trade*. 4 April 2018. Archived (<https://web.archive.org/web/20181012214604/http://www.armstrade.org/includes/periodics/news/2018/0404/115546147/detail.shtml>) from the original on 12 October 2018. Retrieved 25 January 2019.

88. "Замминистра обороны Алексей Криворучко провел рабочее совещание в Ахтубинске" (<http://www.armstrade.org/includes/periodics/news/2018/1012/100049102/detail.shtml>). *armstrade.org*. 12 October 2018. Archived (<https://web.archive.org/web/20181015091735/http://www.armstrade.org/includes/periodics/news/2018/1012/100049102/detail.shtml>) from the original on 15 October 2018. Retrieved 25 January 2019.
89. "Пилотажная группа "Русские Витязи" завершила перевооружение на новые типы авиатехники" (<https://armstrade.org/includes/periodics/news/2020/0710/100558653/detail.shtml>). *armstrade.org* (in Russian). 10 July 2020.
90. Butowski, Piotr; Newdick, Thomas (4 October 2022). "Russian Aggressor Squadron Gets Its First Su-35S Fighter Jets" (<https://www.thedrive.com/the-war-zone/russian-aggressor-squadron-gets-its-first-su-35s-fighter-jets>). *The Drive*. Retrieved 8 October 2022.
91. Butowski *Air International* October 2019, p. 39.
92. de Larrinaga, Nicholas; O'Connor, Sean (3 February 2016). "Russia deploys Su-35S fighters to Syria". *IHS Jane's Defence Weekly*. **53** (13). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).
93. Nikolsky, Alexey (24 May 2017). В боевых действиях в Сирии участвовали практически все летчики российских ВКС (<https://web.archive.org/web/20170525040832/https://www.vedomosti.ru/politics/articles/2017/05/25/691366-boevih-deistviyah-sirii>). *Vedomosti* (in Russian). Archived from the original (<https://www.vedomosti.ru/politics/articles/2017/05/25/691366-boevih-deistviyah-sirii>) on 25 May 2017. Retrieved 26 August 2017.
94. "Российские Су-35 перехватили два турецких F-16, заставив их сбежать" (<http://avia.pro/news/rossiyskie-su-35-perehvatili-dva-tureckih-f-16-zastaviv-ih-sbezhat>). *avia.pro*. 20 August 2019. Retrieved 25 August 2019.
95. "Источники: российские Су-35 экстренно подняты с авиабазы "Хмеймим" для перехвата израильских самолётов" (<http://avia.pro/news/istochniki-rossiyskie-su-35-ekstrenno-podnyaty-s-aviabazy-hmeymim-dlya-perehvata-izraelskih>). *avia.pro*. 10 September 2019. Retrieved 10 September 2019.
96. "Российские истребители заблокировали новую атаку Израиля на Сирию" (<http://avia.pro/news/rossiyskie-istrebiteli-zablokirovali-novuyu-ataku-izrailiya-na-siriyu>). *avia.pro*. 20 September 2019. Retrieved 22 September 2019.
97. "Российские Су-35 перехватили турецкие истребители F-16" (<http://avia.pro/news/rossiyskie-su-35-perehvatili-tureckie-istrebiteli-f-16>). *avia.pro*. 15 October 2019. Retrieved 4 November 2019.
98. "Российские Су-35 вновь перехватили израильские самолёты над Сирией" (<http://avia.pro/news/rossiyskie-su-35-vnov-perehvatili-izraelskie-samolyoty-nad-siriyey>). *avia.pro*. 13 November 2019. Retrieved 1 December 2019.
99. "Новости Сирии 7 декабря 2019 года. Израильским истребителям пришлось бегством спасаться от российских Су-35, Иран доставил в Сирию неизвестные средства ПВО" (<http://avia.pro/blog/novosti-sirii-7-dekabrya-2019-goda-izraelskim-istrebitelyam-prishlos-begstvom-spasatsya-ot>). *avia.pro*. 7 December 2019. Retrieved 21 December 2019.
100. "An unexpected air-to-air battle is raging over Ukraine. Here's what we know about the losses on both sides" (<https://www.businessinsider.com/whats-known-about-russian-and-ukraine-losses-in-aerial-combat-2022-9>). *Insider*. 5 September 2022. "L-39 jet trainer...MiG-29 of 40th TAB piloted by Lt. Col. Yerko Vaycheslav Vladimirvoich...MiG-29 of 40th TAB piloted by Vladimir Kokhansky...Su-27 lost on February 28 on a defensive mission over Koprivnitsky battling Russian fighters...MiG-29 of Maj. Oleksandr Brynzhal, KIA after shot down over Kyiv Oblast in duel...MiG-29 on March 13 downed over Chernihiv...MiG-29 on March 23 piloted by Maj. Dmitry Chumachenko"

101. "An unexpected air-to-air battle is raging over Ukraine. Here's what we know about the losses on both sides" (<https://www.businessinsider.com/whats-known-about-russian-and-ukraine-losses-in-aerial-combat-2022-9>). *Insider*. 5 September 2022. "an Su-35 was recorded attempting to gun down a Ukrainian Mi-14 utility helicopter over Odessa Bay, only to miss. Subsequently, it downed the chopper with a missile, killing a high-ranking naval officer."
102. Ranter, Harro. "Accident Mil Mi-8 , Sunday 6 March 2022" (<https://asn.flightsafety.org/wikibase/298538>). *asn.flightsafety.org*. Retrieved 24 May 2025.
103. Ranter, Harro. "Accident Mil Mi-8 , Sunday 6 March 2022" (<https://asn.flightsafety.org/wikibase/298539>). *asn.flightsafety.org*. Retrieved 24 May 2025.
104. Ranter, Harro. "Accident Sukhoi Su-25 , Sunday 6 March 2022" (<https://asn.flightsafety.org/wikibase/511677>). *asn.flightsafety.org*. Retrieved 24 May 2025.
105. "Датиев Алан Георгиевич" (https://warheroes.ru/hero/hero.asp?Hero_id=34334). *warheroes.ru*. Retrieved 24 May 2025.
106. Cenciotti, David (3 April 2022). "First Russian Su-35S Flanker-E Confirmed Destroyed In Ukraine" (<https://theaviationist.com/2022/04/03/first-russian-su-35s-flanker-e-confirmed-destroyed-in-ukraine/>). *The Aviationist*. Retrieved 4 April 2022.
107. "Малов Сергей Владимирович| Интервью с @Volodymyr Zolkin" (<https://m.youtube.com/watch?v=Nmc38S0ZEzk&feature=youtu.be>). *YouTube*. 5 May 2022.
108. Hambling, David. "Russia May Have Shot Down Another Of Its Warplanes—Were They Aiming At An American Rocket?" (<https://www.forbes.com/sites/davidhambling/2022/07/20/they-did-it-again-russia-shoots-down-another-of-its-own-jets--are-american-rockets-confusing-them/?sh=55ea3e6c4a5a>). *Forbes*.
109. "Apparent Russian Su-35 Shot Down by Ukraine as \$85m Aircraft Goes Up in Flames—Video" (<https://www.newsweek.com/russian-su-35-shot-down-ukraine-aircraft-flames-video-kherson-1726223>). *Newsweek*. 20 July 2022.
110. Malyasov, Dylan (8 February 2023). "Ukrainian Soldiers find wreckage from Russia's high-end Su-35S fighter jet" (<https://defence-blog.com/ukrainian-soldiers-find-wreckage-from-russias-high-end-su-35s-fighter-jet/>). Retrieved 11 May 2023.
111. Axe, David. "Seven Sukhois In Five Days. Ukraine's Patriot Missile Crews Are Shooting Down Russian Jets Faster Than Ever" (<https://www.forbes.com/sites/davidaxe/2024/02/21/s-even-sukhois-in-five-days-ukraines-patriot-missile-crews-are-shooting-down-russian-jets-faster-than-ever/>). *Forbes*. Retrieved 27 May 2024.
112. Axe, David. "Seven Sukhois In Five Days. Ukraine's Patriot Missile Crews Are Shooting Down Russian Jets Faster Than Ever" (<https://www.forbes.com/sites/davidaxe/2024/02/21/s-even-sukhois-in-five-days-ukraines-patriot-missile-crews-are-shooting-down-russian-jets-faster-than-ever/>). *Forbes*. Retrieved 27 May 2024.
113. Ankel, Sophia (4 October 2023). "Russia really did shoot down one of its own prized Su-35 fighter jets by mistake, UK intel says" (<https://www.businessinsider.com/russia-accidentally-owned-own-su-35-fighter-jet-uk-intel-2023-10>). *Business Insider*. Retrieved 27 May 2024.
114. Brennan, David (4 October 2023). "Russia shot down its "most advanced" fighter jet over Tokmak, UK MOD says" (<https://www.newsweek.com/russia-shot-advanced-fighter-jet-tokmak-1831949>). *Newsweek*. Retrieved 27 May 2024.
115. "Ukraine's Defence Forces down 2 more Russian Su-34 jets: three jets downed on 29 February" (<https://www.pravda.com.ua/eng/news/2024/02/29/7444287/>). *Ukrainska Pravda*. 29 February 2024. Retrieved 29 February 2024.
"February is the shortest month of the year, but our sky defenders have achieved the greatest results in downing russian[sic] jets since October 2022" (<https://twitter.com/DefenceU/status/1763195543885324370>). *Ministry of Defense of Ukraine via Twitter*. 29 February 2024. Retrieved 29 February 2024.
116. "Burning Russian 'Combatant' fighter falls over Crimea in flames" (<https://bulgarianmilitary.com/2024/03/28/burning-russian-combatant-fighter-falls-over-crimea-in-flames/>). 28 March 2024. Retrieved 27 May 2024.

117. Reporter, Brendan Cole Senior News (22 May 2023). "Ukraine shoots down Russian Su-35 bomber over Black Sea, air force says" (<https://www.newsweek.com/ukraine-russia-sukhoi-black-sea-1801739>). *Newsweek*. Retrieved 28 May 2024.
118. "Moscow's Aerospace Forces: No air of superiority" (<https://www.iiss.org/en/online-analysis/military-balance/2024/02/moscows-aerospace-forces-no-air-of-superiority/>). *I/SS*. Retrieved 27 May 2024.
119. Axe, David. "Seven Sukhois In Five Days. Ukraine's Patriot Missile Crews Are Shooting Down Russian Jets Faster Than Ever" (<https://www.forbes.com/sites/davidaxe/2024/02/21/s-even-sukhois-in-five-days-ukraines-patriot-missile-crews-are-shooting-down-russian-jets-faster-than-ever/>). *Forbes*. Retrieved 27 May 2024.
120. Newdick, Thomas (9 June 2025). "Claims Swirl Around Russian Su-35S Flanker Shootdown" (<https://www.twz.com/air/claims-swirl-around-russian-su-35s-flanker-shootdown>). *The War Zone*. Retrieved 3 July 2025.
121. Vladyslava Kovalenko. "Video emerges of Su-35 fighter jet shot down over Russia" (<https://newsukraine.rbc.ua/news/video-emerges-of-su-35-fighter-jet-shot-down-1749292844.html>). *RBC Ukraine*. Retrieved 7 June 2025.
122. Gill and Kim 1995, p. 60.
123. Zhao 2004, p. 216.
124. "Russia inks contract with China on Su-35 deliveries" (<https://web.archive.org/web/20151120184705/http://tass.ru/en/defense/837662>). TASS. 19 November 2015. Archived from the original (<http://tass.ru/en/defense/837662>) on 20 November 2015. Retrieved 20 November 2015.
125. Stolyarov, Gleb (19 November 2015). "UPDATE 2-Russia, China sign contract worth over \$2 bln for Su-35 fighter jets -source" (<https://web.archive.org/web/20151120154318/https://www.reuters.com/article/2015/11/20/russia-china-jets-idUSL3N13F1FP20151120>). *Reuters*. Archived from the original (<https://www.reuters.com/article/russia-china-jets-idUSL3N13F1FP20151120#keQBH8RmK81myGve.97>) on 20 November 2015. Retrieved 20 November 2015.
126. Domínguez, Gabriel (24 November 2015). "Russia needs China to buy its weapons" (<https://www.dw.com/en/why-russia-needs-china-to-buy-its-weapons/a-18870472>). *Deutsche Welle*.
127. Minnick, Wendell (20 November 2015). "Russia-China Su-35 Deal Raises Reverse Engineering Issue" (<https://www.defensenews.com/air/2015/11/20/russia-china-su-35-deal-raises-reverse-engineering-issue/>). *Defense News*. Retrieved 29 July 2017.
128. "Russia sells Su-35s to China" (<https://www.upi.com/Defense-News/2010/11/23/Russia-sells-Su-35s-to-China/71451290548527/>). *United Press*. 23 November 2010.
129. Wood, Peter (10 October 2013). "Why China Wants the Su-35" (<https://jamestown.org/program/why-china-wants-the-su-35/>). *Jamestown*.
130. Johnson, Reuben (25 November 2015). "Su-35 deal signals PLAAF's lack of faith in Chinese defence sector". *IHS Jane's Defence Weekly*. **53** (3). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).
131. Choi, Chi-yuk (26 March 2013). "China to buy Lada-class subs, Su-35 fighters from Russia" (<https://web.archive.org/web/20130326183826/http://www.scmp.com/news/china/article/1199448/china-buy-russian-fighters-submarines>). *South China Morning Post*. Archived from the original (<https://www.scmp.com/news/china/article/1199448/china-buy-russian-fighters-submarines>) on 26 March 2013. Retrieved 26 March 2013.
132. Johnson, Reuben (11 September 2013). "Reports of Su-35 sale to China met with scepticism". *IHS Jane's Defence Weekly*. **50** (41). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).
133. Johnson, Reuben (14 March 2012). "Russian industry wary of Su-35 sale to China". *IHS Jane's Defence Weekly*. **49** (14). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).

134. Pyadushkin 2014, p. 19.
135. Sloman, Jesse; Dickey, Lauren (1 June 2015). "Why China's Air Force Needs Russia's SU-35" (<https://web.archive.org/web/20170702062341/https://thediplomat.com/2015/06/why-chinas-air-force-needs-the-su-35/>). *The Diplomat*. Archived from the original (<https://thediplomat.com/2015/06/why-chinas-air-force-needs-the-su-35/>) on 2 July 2017. Retrieved 30 July 2017.
136. Karnozov, Vladimir (14 December 2012). "Sukhoi Nears Deal with China on Su-35 Fighters" (<https://web.archive.org/web/20170729182904/http://www.ainonline.com/aviation-news/defense/2012-12-14/sukhoi-nears-deal-china-su-35-fighters>). *AIN Online*. Archived from the original (<http://www.ainonline.com/aviation-news/defense/2012-12-14/sukhoi-nears-deal-china-su-35-fighters>) on 29 July 2017. Retrieved 29 July 2017.
137. "PLA news portal: Su-35 intended to be last type of imported fighter" (<http://en.people.cn/n3/2016/1230/c90000-9161327.html>). *People's Daily*. 30 December 2016. Archived (<https://web.archive.org/web/20180720065706/http://en.people.cn/n3/2016/1230/c90000-9161327.html>) from the original on 20 July 2018. Retrieved 21 September 2018.
138. Karnozov, Vladimir (27 December 2018). "China Accepts Last Batch of Su-35s, Test Fires S-400" (<https://www.ainonline.com/aviation-news/defense/2018-12-27/china-accepts-last-batch-su-35s-test-fires-s-400>). *AIN Online*. Archived (<https://web.archive.org/web/20190221112136/https://www.ainonline.com/aviation-news/defense/2018-12-27/china-accepts-last-batch-su-35s-test-fires-s-400>) from the original on 21 February 2019. Retrieved 21 February 2019.
139. Zhao, Lei (6 January 2017). "Air Force receives 4 of Russia's latest fighters" (https://web.archive.org/web/20170812101142/http://www.chinadaily.com.cn/china/2017-01/06/content_27877527.htm). *China Daily*. Archived from the original (http://www.chinadaily.com.cn/china/2017-01/06/content_27877527.htm) on 12 August 2017. Retrieved 12 August 2017.
140. "China Takes Delivery of 10 Russian Su-35 Fighter Jets" (<https://thediplomat.com/2018/01/china-takes-delivery-of-10-russian-su-35-fighter-jets/>). *The Diplomat*. 4 January 2018. Retrieved 23 February 2020.
141. "Su-35S fighter jets officially entered service with PLAAF" (<http://www.airrecognition.com/index.php/archive-world-worldwide-news-air-force-aviation-aerospace-air-military-defence-industry/global-defense-security-news/global-news-2018/may/4272-su-35s-fighter-jets-officially-entered-service-with-plaaf.html>). *www.airrecognition.com*. Archived (<https://web.archive.org/web/20180921153435/http://www.airrecognition.com/index.php/archive-world-worldwide-news-air-force-aviation-aerospace-air-military-defence-industry/global-defense-security-news/global-news-2018/may/4272-su-35s-fighter-jets-officially-entered-service-with-plaaf.html>) from the original on 21 September 2018. Retrieved 4 January 2019.
142. "Russia Offers China Another Batch of Su-35 Fighter Jets" (<https://thediplomat.com/2019/06/russia-offers-china-another-batch-of-su-35-fighter-jets/>). *The Diplomat*. 27 June 2019. Retrieved 16 July 2019.
143. Honrada, Gabriel (16 April 2022). "China's J-20 fighters begin South China Sea patrols" (<https://asiatimes.com/2022/04/chinas-j-20-fighters-begin-south-china-sea-patrols/>). *Asia Times*.
144. "U.S. sanctions China for buying Russian fighter jets, missiles" (<https://www.reuters.com/article/us-usa-russia-sanctions/us-sanctions-china-for-buying-russian-fighter-jets-missiles-idUSKCN1M02TP>). *Reuters*. 20 September 2018. Archived (<https://web.archive.org/web/20190710092457/https://www.reuters.com/article/us-usa-russia-sanctions/us-sanctions-china-for-buying-russian-fighter-jets-missiles-idUSKCN1M02TP>) from the original on 10 July 2019. Retrieved 16 July 2019.
145. "Russia trains Iranian pilots ahead of Su-35 transfer – report" (<https://www.aerotime.aero/articles/russia-trains-iranian-pilots-ahead-of-su-35-transfer-report>). *aerotime.aero*. 12 December 2022.
146. "Russia trains Iranian pilots on Su-35 fighters" (<https://mil.in.ua/en/news/russia-trains-iranian-pilots-on-su-35-fighters/>). *mil.in.ua*. 10 December 2022.

147. "Iran Air Force to get 24 Sukhoi Su-35 Flanker-E fighter jets in Spring 2023" (<https://www.airrecognition.com/index.php/news/defense-aviation-news/2023-news-aviation-aerospace/january/8867-iran-air-force-to-get-24-sukhoi-su-35-fighter-jets-in-spring-2023.html>). *www.airrecognition.com*.
148. "Iran Implies Collapse Of Deal To Buy Russia's Su-35 Warplanes" (<https://www.iranintl.com/en/202307203141>). *Iran International*. 20 July 2023. Retrieved 21 July 2023.
149. "Iran finalises deal to buy Russian fighter jets - Tasnim" (<https://www.reuters.com/world/iran-finalises-deal-buy-russian-fighter-jets-tasnim-2023-11-28/>). *Reuters*. 28 November 2023.
150. Iddon, Paul (20 April 2024). "Iran Might Receive Its First Su-35 Flanker Fighters From Russia Next Week" (<https://www.forbes.com/sites/pauliddon/2024/04/20/iran-might-receive-its-first-su-35-flanker-fighters-from-russia-next-week/>). *Forbes*.
151. dpa (21 April 2024). "Iran Disputes Reports Of First Delivery Of Russian Su-35 Fighter Jets" (<https://www.rferl.org/a/iran-disputes-reports-delivery-russian-fighter-jets/32914490.html>). *RadioFree Europe / Radio Liberty*.
152. "Iran's Revolutionary Guards commander says Iran purchased Russian-made Sukhoi 35 fighter jets" (<https://www.reuters.com/business/aerospace-defense/irans-revolutionary-guard-s-commander-says-iran-purchased-russian-made-sukhoi-35-2025-01-27/>). *Reuters*. 27 January 2025.
153. Daftari, Amir (24 September 2025). "Russia Delivers MiG-29 Jets to Iran Air Force" (<https://www.newsweek.com/russia-delivers-mig-29-jets-iran-air-force-10479982>). *Newsweek*. Retrieved 25 September 2025.
154. Osborne, Tony (21 February 2017). "Russia, UAE To Partner On Fighter Development" (<http://aviationweek.com/defense/russia-uae-partner-fighter-development>). *Aerospace Daily & Defense Report*. Aviation Week. Archived (<https://web.archive.org/web/20170909102030/http://aviationweek.com/defense/russia-uae-partner-fighter-development>) from the original on 9 September 2017. Retrieved 9 September 2017.
155. Yousef, Deena Kamel; Jasper, Christopher. "U.A.E. to Build Russian Warplane as Mideast Tensions Rise" (<https://web.archive.org/web/20170319183720/https://www.bloomberg.com/news/articles/2017-02-20/u-a-e-to-build-russian-warplane-as-iran-stokes-mideast-tensions>). Bloomberg. Archived from the original (<https://www.bloomberg.com/news/articles/2017-02-20/u-a-e-to-build-russian-warplane-as-iran-stokes-mideast-tensions>) on 19 March 2017. Retrieved 19 August 2017.
156. Diplomat, Abraham Ait, The. "Could Russia Design a Fifth-Generation Variant of the Su-35 for India?" (<https://thedi diplomat.com/2018/02/could-russia-design-a-fifth-generation-variant-of-the-su-35-for-india/>). *The Diplomat*. Archived (<https://web.archive.org/web/20190326145235/https://thedi diplomat.com/2018/02/could-russia-design-a-fifth-generation-variant-of-the-su-35-for-india/>) from the original on 26 March 2019. Retrieved 4 January 2019.
157. "MMRCA 2.0 Contenders" (<http://sps-aviation.com/story/?id=2370&h=MMRCA-20-Contenders>). *sps-aviation.com*.
158. Safranov, Ivan (4 March 2016). "Без кредита не взлететь" (<https://web.archive.org/web/20170711005800/https://www.kommersant.ru/doc/2931872>). *Kommersant* (in Russian). Archived from the original (<https://www.kommersant.ru/doc/2931872>) on 11 July 2017. Retrieved 26 June 2017.
159. "L'Algérie serait le premier client export pour le Su-57 et le Su-34" (<https://www.menadefense.net/algerie/lalgerie-serait-le-premier-client-export-pour-le-su-57-et-le-su-34/>). *menadefense.net*. 27 December 2019. Retrieved 31 December 2019.
160. "US removes Turkey from F-35 fighter jet programme" (<https://www.bbc.com/news/world-us-canada-49023115>). *BBC*. 17 July 2019. Retrieved 18 July 2019.
161. "Russia Offers Sukhoi Su-35 to Turkey after US Removes from F-35 Programme" (<https://www.news18.com/news/auto/russia-offers-sukhoi-su-35-to-turkey-after-us-removes-from-f-35-programme-2236287.html>). *News18.com*. 18 July 2019. Retrieved 18 July 2019.

162. "Russia 'ready to cooperate' to sell Turkey Su-57 fighter jets" (<http://www.hurriyetdailynews.com/russia-ready-to-cooperate-to-sell-turkey-su-57-fighter-jets-143123>). *hurriyetdailynews.com*. Hürriyet Daily News. 3 May 2019. Archived (<https://web.archive.org/web/20190504135930/http://www.hurriyetdailynews.com/russia-ready-to-cooperate-to-sell-turkey-su-57-fighter-jets-143123>) from the original on 4 May 2019. Retrieved 6 May 2019.
163. Cave, Damien (27 October 2025). "Is the U.S. Losing in Vietnam? Russia, North Korea and China Are Gaining" (<https://www.nytimes.com/2025/10/27/world/asia/vietnam-russia-relations.html>). *The New York Times*. Archived (<https://web.archive.org/web/20251028001054/http://www.nytimes.com/2025/10/27/world/asia/vietnam-russia-relations.html>) from the original on 28 October 2025. Retrieved 28 October 2025. "Previously unreported, the air-defense orders suggested to analysts — who confirmed that the documents offered a credible glimpse of Vietnam's plans — that a deal for Su-35s and other aircraft was probably already done. One of several Vietnamese officials who confirmed large orders from Russia said the country purchased 40 new Su-35 and Su-30 fighter jets, as part of a deal worth \$8 billion..."
164. Litovkin, Nikolai; Egorova, Kira (27 June 2016). "Russian aircraft sales boosted by Syrian campaign" (https://rbth.com/defence/2016/06/27/russian-aircraft-sales-boosted-by-syrian-campaign_606547). *Russia Beyond the Headlines*. Archived (https://web.archive.org/web/20170329102132/http://rbth.com/defence/2016/06/27/russian-aircraft-sales-boosted-by-syrian-campaign_606547) from the original on 29 March 2017. Retrieved 16 November 2016.
165. Nkala, Oscar (3 May 2016). "Algeria to test Su-35 fighter ahead of possible acquisition" (http://web.archive.org/web/20170808073229/http://www.defenceweb.co.za/index.php?option=com_content&view=article&id=43349). *Defenceweb.co.za*. Archived from the original (http://www.defenceweb.co.za/index.php?option=com_content&view=article&id=43349) on 8 August 2017. Retrieved 26 June 2017.
166. "Arms Trade; Kazakh Defense Ministry interested in buying batch of Su-35 fighter jets". *Russia & CIS Defense Industry Weekly*. Interfax. 3 June 2016.
167. "North Korea asked Russia for Su-35 fighter jet supplies – reports" (<https://web.archive.org/web/20170217173134/http://tass.com/world/770538>). TASS. 9 January 2015. Archived from the original (<http://tass.com/world/770538>) on 17 February 2017. Retrieved 20 November 2015.
168. "Sudan seeking Su-35s" (https://archive.today/20171201231440/http://www.defenceweb.co.za/index.php?option=com_content&view=article&id=50032:sudan-seeking-su-35s&catid=35:Aerospace&Itemid=107). *Defence Web*. 29 November 2017. Archived from the original (http://www.defenceweb.co.za/index.php?option=com_content&view=article&id=50032:sudan-seeking-su-35s&catid=35:Aerospace&Itemid=107) on 1 December 2017. Retrieved 1 December 2017.
169. Lantratov, Constantine (31 March 2004). "'Сухой' против EmBrAer" (<https://web.archive.org/web/20131029193239/http://www.kommersant.ru/doc/462335>). *Kommersant* (in Russian). Archived from the original (<http://www.kommersant.ru/doc/462335>) on 29 October 2013. Retrieved 19 April 2013.
170. Rivers 2005, p. 18.
171. Lantratov, Constantine (28 February 2005). Бразилии не нужны новые истребители (<http://web.archive.org/web/20130824062002/http://www.kommersant.ru/doc/550698>). *Kommersant* (in Russian). Archived from the original (<http://www.kommersant.ru/doc/550698>) on 24 August 2013. Retrieved 20 April 2013.
172. Komarov 2002, p. 61.
173. Trimble, Stephen (6 October 2008). "Brazil names three finalists for F-X2 contract, rejects three others" (<https://web.archive.org/web/20121002082836/http://www.flightglobal.com/news/articles/brazil-names-three-finalists-for-f-x2-contract-rejects-three-316814/>). *Flightglobal*. Archived from the original (<http://www.flightglobal.com/news/articles/brazil-names-three-finalists-for-f-x2-contract-rejects-three-316814/>) on 2 October 2012. Retrieved 24 August 2013.

174. "Entrevista com Anatoly Isaikin Diretor General da EEFU Rosoboronexport sobre o Programa F-X2" (<https://web.archive.org/web/20131029193416/http://www.defesanet.com.br/fx2/noticia/187/Entrevista-com-Anatoly-Isaikin-Diretor-General-da-EEFU-Rosoboronexport-sobre-o-Programa-F-X2>). *DefesaNet* (in Portuguese). 6 October 2009. Archived from the original (<http://www.defesanet.com.br/fx2/noticia/187/Entrevista-com-Anatoly-Isaikin-Diretor-General-da-EEFU-Rosoboronexport-sobre-o-Programa-F-X2>) on 29 October 2013. Retrieved 27 March 2013.
175. "Russia to pitch T-50 for Brazilian fighter deal" (<https://web.archive.org/web/20131016091901/https://www.flightglobal.com/news/articles/russia-to-pitch-t-50-for-brazilian-fighter-deal-391737/>). *Flightglobal*. 16 October 2016. Archived from the original (<https://www.flightglobal.com/news/articles/russia-to-pitch-t-50-for-brazilian-fighter-deal-391737/>) on 16 October 2013. Retrieved 12 September 2017.
176. Salles, Filipe (18 December 2013). "Saab wins Brazil's F-X2 fighter contest with Gripen NG" (<https://web.archive.org/web/20140112053200/https://www.flightglobal.com/news/articles/saab-wins-brazil39s-f-x2-fighter-contest-with-gripen-394289/>). *Flightglobal*. Archived from the original (<https://www.flightglobal.com/news/articles/saab-wins-brazil39s-f-x2-fighter-contest-with-gripen-394289/>) on 12 January 2014. Retrieved 9 September 2017.
177. Binnie, Jeremy (18 March 2019). "Egypt reportedly orders Su-35 fighters" (<https://www.janes.com/article/87282/egypt-reportedly-orders-su-35-fighters>). *Jane's* 360. London. Archived (<https://web.archive.org/web/20190318224216/https://www.janes.com/article/87282/egypt-reportedly-orders-su-35-fighters>) from the original on 18 March 2019. Retrieved 18 March 2019.
178. Сафронов, Иван; Джорджевич, Александра (18 March 2019). "Су-35 укрепят египетскую силу" (<https://www.kommersant.ru/doc/3915483>). *Kommersant*. Archived (<https://web.archive.org/web/20190318131445/https://www.kommersant.ru/doc/3915483>) from the original on 18 March 2019. Retrieved 18 March 2019.
179. " "Рособоронэкспорт" опроверг наличие контракта на поставку Су-35 Египту" (<https://www.interfax.ru/russia/655788>). *Interfax*. 26 March 2019. Archived (<https://web.archive.org/web/20190605020831/https://www.interfax.ru/russia/655788>) from the original on 5 June 2019. Retrieved 17 June 2019.
180. "Pompeo: Egypt would face sanctions over Russian Su-35s" (<https://www.aa.com.tr/en/americas/pompeo-egypt-would-face-sanctions-over-russian-su-35s/1447386>). *Anadolu Agency*. 10 April 2019. Retrieved 16 July 2019.
181. "Russia begins production of Egyptian Su-35s" (<https://www.defenceweb.co.za/aerospace/aerospace-aerospace/russia-begins-production-of-egyptian-su-35s/>). *defenceWeb*. 19 May 2020. Retrieved 22 May 2020.
182. "Here Are The First Photos Of Egypt's New Su-35 "Super Flankers" " (<https://www.thedrive.com/the-war-zone/35055/here-are-the-first-photos-of-egypts-new-su-35-super-flankers>). *The War Zone*. 23 July 2020. Retrieved 26 June 2021.
183. International, Flight; Embraer, in association with. "World Air Forces 2021" (<https://www.flightglobal.com/reports/world-air-forces-2021/141456.article>). *Flight Global*. Retrieved 15 February 2021.
184. "U.S. Plans to sell F-15 fighters to Egypt amid human rights dispute" (<https://www.politico.com/news/2022/03/15/f-15-fighter-jets-sale-egypt-00017339>). *Politico*. 15 March 2022.
185. "Iran buys 24 Sukhoi Su-35 Flanker-E fighters produced for Egypt" (<https://bulgarianmilitary.com/2022/09/04/iran-buys-24-sukhoi-su-35-flanker-e-fighters-produced-for-egypt/>). 4 September 2022.
186. "Egypt's Sukhoi Su-35Es sold to Iran due to US pressure" (<https://www.military.africa/2022/09/egypts-sukhoi-su-35es-sold-to-iran-due-to-us-pressure/>). 5 September 2022.
187. "US to supply F-15s to Egypt" (<https://www.egyptdefenceexpo.com/news/us-to-supply-f-15s-to-egypt>).

188. Grevatt, Jon (5 November 2014). "IndoDefence 2014: UAC announces Su-35 bid for Indonesian fighter competition". *IHS Jane's Defence Weekly*. **51** (50). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).
189. "Kemhan Akan Ganti F-5 dengan Sukhoi SU-35" (<https://web.archive.org/web/20170807232757/http://www.beritasatu.com/nasional/303974-kemhan-akan-ganti-f5-dengan-sukhoi-su35.html>). *BeritaSatu* (in Indonesian). 2 September 2015. Archived from the original (<http://www.beritasatu.com/nasional/303974-kemhan-akan-ganti-f5-dengan-sukhoi-su35.html>) on 7 August 2017. Retrieved 12 August 2017.
190. Grevatt, Jon (2 September 2015). "Indonesia selects Su-35 to meet air combat requirement". *IHS Jane's Defence Weekly*. **52** (42). Coulsdon, UK: Jane's Information Group. ISSN 2048-3430 (<https://search.worldcat.org/issn/2048-3430>).
191. Grevatt, Jon (15 February 2018). "Indonesia finalises contract to procure Su-35 fighter aircraft" (<https://web.archive.org/web/20180215144525/http://www.janes.com/article/77928/indonesia-finalises-contract-to-procure-su-35-fighter-aircraft>). *IHS Jane's* 360. Bangkok. Archived from the original (<http://www.janes.com/article/77928/indonesia-finalises-contract-to-procure-su-35-fighter-aircraft>) on 15 February 2018. Retrieved 15 February 2018.
192. "Trump Threat Spurred Indonesia to Drop Russia, China Arms Deals" (<https://www.bloomberg.com/news/articles/2020-03-12/trump-threat-spurred-indonesia-to-drop-russia-china-arms-deals>). *bloomberg.com*. 12 March 2020. Retrieved 13 April 2020.
193. Grevatt, Jon (18 March 2020). "Update: Indonesia points to F-35 ambitions" (<https://www.janes.com/defence-news/news-detail/update-indonesia-points-to-f-35-ambitions>). *janes.com*. Retrieved 20 June 2021.
194. "Proyek Pembelian Sukhoi Su-35 Masih Lanjut" (<https://web.archive.org/web/20210131035826/https://fin.co.id/2020/07/09/proyek-pembelian-sukhoi-su-35-masih-lanjut/>). Archived from the original (<https://fin.co.id/2020/07/09/proyek-pembelian-sukhoi-su-35-masih-lanjut/>) on 31 January 2021.
195. antaranews.com (22 December 2021). "TNI AU pastikan beli pesawat tempur Rafale dan F-15 EX" (<https://www.antaranews.com/berita/2600597/tni-au-pastikan-beli-pesawat-tempur-rafale-dan-f-15-ex>). *Antara News*. Retrieved 22 December 2021.
196. "Tanggapan Terhadap Pemberitaan di Media Tentang Pengadaan Pesawat Mirage 2000-5" (<https://www.kemhan.go.id/2023/06/15/tanggapan-terhadap-pemberitaan-di-media-tentang-pengadaan-pesawat-mirage-2000-5.html>) [Responses to Media Coverage of Procurement of Mirage 2000-5 Aircraft] (Press release) (in Indonesian). Indonesian Ministry of Defence. 14 June 2023. Retrieved 15 June 2023.
197. Abrams, A. B. (20 May 2024). "Russian Fighter Deal Reflects Indonesia's Goal of a 'Sanctions-Proof' Economy" (<https://thediplomat.com/2024/05/russian-fighter-deal-reflects-indonesias-goal-of-a-sanctions-proof-economy/>). *The Diplomat*.
198. Sherman 2001, p. 22.
199. Joo and Kwak 2001, p. 205.
200. Gethin 1998, p. 32.
201. Govindasamy, Siva (22 October 2007). "South Korean F-15K deal may close by end 2007" (<https://web.archive.org/web/20121105100722/http://www.flightglobal.com/news/articles/south-korean-f-15k-deal-may-close-by-end-2007-218740/>). *Flightglobal*. Archived from the original (<http://www.flightglobal.com/news/articles/south-korean-f-15k-deal-may-close-by-end-2007-218740/>) on 5 November 2012. Retrieved 26 March 2013.
202. Mader 2007, p. 102.
203. Mader 2007, p. 103.

204. Sweeney, Jack (15 October 2008). "Venezuela buys Russian aircraft, tanks to boost power" (https://web.archive.org/web/20100529022832/http://www.upi.com/Business_News/Security-Industry/2008/10/15/Venezuela-buys-Russian-aircraft-tanks-to-boost-power/UPI-11881224089163/). *United Press International*. Archived from the original (http://www.upi.com/Business_News/Security-Industry/2008/10/15/Venezuela-buys-Russian-aircraft-tanks-to-boost-power/UPI-11881224089163/) on 29 May 2010. Retrieved 26 March 2013.
205. "Russia big loser in Arab arms market slump" (https://web.archive.org/web/20110307050119/http://www.upi.com/Business_News/Security-Industry/2011/03/04/Russia-big-loser-in-Arab-arms-market-slump/UPI-22991299266527/). *United Press International*. 4 March 2011. Archived from the original (http://www.upi.com/Business_News/Security-Industry/2011/03/04/Russia-big-loser-in-Arab-arms-market-slump/UPI-22991299266527/) on 7 March 2011. Retrieved 27 March 2013.
206. "Σαράντα οκτώ Su-35 παραδίδονται το 2015 – Υποψήφιο στην αξιολόγηση της Π.Α – Video: Su-35 Vs EF-2000" (<https://web.archive.org/web/20130919062412/http://www.defencenet.gr/defence/o/14398>). *Defencenet.gr* (in Greek). 20 September 2010. Archived from the original (<http://www.defencenet.gr/defence/o/14398>) on 19 September 2013. Retrieved 8 April 2013.
207. Andrews 2003, p. 57.
208. Gordon 2007, p. 129.
209. Gordon 2007, pp. 129, 591.
210. Gordon 2007, pp. 151, 154, 158.
211. Gordon 2007, pp. 170–171.
212. Andrews 2003, p. 59.
213. Gordon 2007, p. 172.
214. Trimble, Stephen (20 August 2009). "Russia signs \$2.5 billion deal for 64 Sukhoi fighters" (<https://web.archive.org/web/20130824023307/http://www.flightglobal.com/news/articles/russia-signs-25-billion-deal-for-64-sukhoi-fighters-331265/>). *Flightglobal*. Archived from the original (<http://www.flightglobal.com/news/articles/russia-signs-25-billion-deal-for-64-sukhoi-fighters-331265/>) on 24 August 2013. Retrieved 18 July 2010.
215. admin (26 June 2023). "Rostec State Corporation Delivers New Sukhoi Su-35S Fighter Jets to Russian Air Forces" (<https://militaryleak.com/2023/06/26/rostec-state-corporation-delivers-new-sukhoi-su-35s-fighter-jets-to-russian-air-forces/?amp=1>). *MilitaryLeak*. Retrieved 6 July 2023.
216. "Chapter Five: Russia and Eurasia" (<https://www.tandfonline.com/doi/full/10.1080/04597222.2020.1707966>). *The Military Balance*. **120** (1): 166–219. 1 January 2020. doi:10.1080/04597222.2020.1707966 (<https://doi.org/10.1080%2F04597222.2020.1707966>). ISSN 0459-7222 (<https://search.worldcat.org/issn/0459-7222>). S2CID 219627993 (<https://api.semanticscholar.org/CorpusID:219627993>).
217. "The Su-35SM will use an AESA radar and a Himalayan EW system" (<https://www.airspace-review.com/2023/01/13/su-35sm-akan-menggunakan-radar-aesa-dan-sistem-ew-himalaya/>). 13 January 2023.
218. Akramov (13 March 2025). "Premier vol d'un SU 35 de l'armée de l'air Algérienne" (<https://www.menadefense.net/premier-vol-dun-su-35-de-larmee-de-lair-algerienne/>). *Menadefense*. Retrieved 13 March 2025.
219. admin (14 March 2025). "Algeria Inducts Sukhoi Su-35: Strengthening Air Dominance Amid Regional Power Shifts" (<https://defencesecurityasia.com/en/algeria-inducts-su-35-strengthening-air-dominance-amid-regional-power-shifts/>). *Defence Security Asia*. Retrieved 15 March 2025.
220. "Tehran purchases Russian Su-35 fighter jets — Iranian general" (<https://tass.com/defense/1904833>). *TASS*. 27 January 2025. Retrieved 11 November 2025.

221. "Iran's Revolutionary Guards commander says Iran purchased Russian-made Sukhoi 35 fighter jets" (<https://www.reuters.com/business/aerospace-defense/irans-revolutionary-guard-s-commander-says-iran-purchased-russian-made-sukhoi-35-2025-01-27/>). *Reuters*. 27 January 2025. Retrieved 11 November 2025.
222. "Почему ВВС России в 2019 году получили мало новой техники?" (<https://regnum.ru/new/s/polit/2827280.html>). *regnum.ru*. 20 January 2020. Retrieved 26 June 2021.
223. "Поставки самолётов в Вооружённые Силы России в 2021 году. Часть 1" (<https://www.soldat.ru/news/1540.html>). *www.soldat.ru*.
224. "ОАК передала самолеты Су-35С для ВКС России" (<https://www.uacrussia.ru/ru/press-center/news/oak-peredala-samolety-su-35s-dlya-vks-rossii>).
225. "ЦАМТО / / Минобороны России передана очередная партия многофункциональных истребителей Су-35С" (<https://armstrade.org/includes/periodics/news/2022/1205/094070653/detail.shtml>). *armstrade.org*.
226. "Russian Aerospace Forces have taken delivery of some new Su-35S fighter jets — minister" (<https://tass.com/defense/1637325>).
227. "Defense contractor delivers 2nd batch of Su-35S fighters to Russian troops" (<https://tass.com/defense/1646263>).
228. Dubois, Gastón (28 September 2023). "The Russian Air Force received new batches of Su-57 and Su-35S fighters" (<https://www.aviacionline.com/2023/09/the-russian-air-force-received-new-batches-of-su-57-and-su-35s-fighters/>). *Aviacionline - Últimas noticias de aviación en Latinoamérica y el Caribe, España y el mundo*.
229. "Su-35 & JAS-39: Russian Air Force Receives New Batch Of Super Flankers; Saab To Upgrade Its Gripen C/D Fighters" (<https://www.eurasiantimes.com/su-35-saab-to-upgrade-it-s-gripen-c-d-fighters/>). 24 October 2023.
230. "ЦАМТО / / ОАК передала Минобороны очередную партию истребителей Су-35С" (<https://armstrade.org/includes/periodics/news/2023/1124/105076587/detail.shtml>). *ЦАМТО / Центр анализа мировой торговли оружием* (in Russian). 24 November 2023. Retrieved 26 November 2023.
231. Максимов, Илья (12 April 2024). "Первую в этом году партию истребителей Су-35С передали в войска" (<https://rg.ru/2024/04/12/pervuiu-v-etom-godu-partiiu-istrebitelej-su-35s-peredali-v-vojska.html>). *Российская газета* (in Russian). Retrieved 13 April 2024.
232. "ОАК передала Минобороны новые самолеты Су-35С" (<https://rostec.ru/news/oak-peredala-minoborony-novye-samolety-su-35s/>). *Ростех* (in Russian). Retrieved 9 May 2024.
233. Suci, Peter (11 November 2024). "NATO Is Freaked: Russia Is Getting New Su-57 and Su-35 Fighter Jets" (<https://nationalinterest.org/blog/buzz/nato-freaked-russia-getting-new-su-57-and-su-35-fighter-jets-213665>). *The National Interest*. Retrieved 12 November 2024.
234. "Russian Air Force Receives New Su-35 Fighters: Growing Exports Likely to Reduce Domestic Procurement" (<https://militarywatchmagazine.com/article/russian-air-force-receives-new-su35-fighters-growing-exports>). *Military Watch Magazine*. 30 March 2025. Retrieved 31 March 2025.
235. "Russian Air Force Receives New Su-35 Fighters Amid Production Scale Expansion" (<https://militarywatchmagazine.com/article/russian-air-force-su35-production-scale>). *Military Watch Magazine*. 13 May 2025. Retrieved 14 May 2025.
236. "ОАК передала ВКС России очередную партию новых истребителей Су-35С" (<https://rostec.ru/media/news/oak-peredala-vks-rossii-ocherednuyu-partiyu-novykh-istrebiteley-su-35s/>). *Ростех* (in Russian). Retrieved 26 June 2025.
237. Malyasov, Dylan (21 August 2025). "Russia receives new batch of Su-35 fighter jets" (<https://defence-blog.com/russia-receives-new-batch-of-su-35-fighter-jets/>). *Defence Blog*. Retrieved 22 August 2025.

238. Malyasov, Dylan (24 September 2025). "Russian military receives new Su-35 fighter jets" (<https://defence-blog.com/russian-military-receives-new-su-35-fighter-jets/>). *Defence Blog*. Retrieved 25 September 2025.
239. Malyasov, Dylan (1 November 2025). "Russia adds newly built Su-35S jets to its air fleet" (<https://defence-blog.com/russia-adds-newly-built-su-35s-jets-to-its-air-fleet/?amp>). *Defence Blog – Military and Defense News*. Retrieved 3 November 2025.
240. "Russian Defense Ministry, Rostec sign contracts for new military hardware worth over 300 bln rubles" (<https://interfax.com/newsroom/top-stories/69681/>). *interfax.com*. 25 August 2020. Retrieved 26 June 2021.
241. "Минфин и "Роскосмос" вступили в публичные препирательства о сокращении расходов" (<https://www.vedomosti.ru/politics/articles/2020/08/25/837739-minfin-roskosmos>). *Ведомости*. 26 August 2020. Retrieved 26 June 2021.
242. "Air regiment in central Russia rearmed with latest Su-35 fighter jets" (<https://tass.com/defense/1071208>). *TASS*. 31 July 2019. Retrieved 3 August 2019.
243. "Три многоцелевых истребителя Су-35 поступили в авиаполк ЗВО в Тверской области" (<http://www.armstrade.org/includes/periodics/news/2019/1017/101554875/detail.shtml>). *armstrade.org*. 17 October 2019. Retrieved 11 November 2019.
244. "ЦАМТО / / В Астраханский Центр ВКС России впервые поступили истребители Су-35С" (<https://armstrade.org/includes/periodics/news/2022/0912/093069439/detail.shtml>).
245. "Russia increases operational jets in Syria after US missile strike [+ Photos]" (<https://web.archive.org/web/20180502012315/https://www.almasdarnews.com/article/russia-increases-operational-jets-in-syria-after-us-missile-strike-photos/>). *Al-Masdar News*. 30 April 2018. Archived from the original (<https://www.almasdarnews.com/article/russia-increases-operational-jets-in-syria-after-us-missile-strike-photos/>) on 2 May 2018. Retrieved 1 May 2018.
246. "Russian Su-35S fighter crashes, pilot ejects safely" (http://www.xinhuanet.com/english/2021-08/01/c_1310099640.htm). *xinhuanet.com*. Retrieved 2 August 2021.
247. Rodionov, Maxim (28 March 2024). "Russian military plane crashes in sea off Crimea - local governor" (<https://www.reuters.com/world/europe/russian-military-plane-crashes-sea-off-crimea-local-governor-2024-03-28/>). *Reuters*.
248. "Su-35" (<https://web.archive.org/web/20120730185357/http://www.knaapo.ru/eng/products/su-35/index.wbp>). КНААПО. Archived from the original (<http://www.knaapo.ru/eng/products/su-35/index.wbp>) on 30 July 2012. Retrieved 25 April 2013.
249. "Sukhoi Su-35", *Jane's All the World's Aircraft*. Jane's Information Group. 6 February 2013.
250. Butowski *Air International* October 2019, p. 38.
251. "В интересах ГПВ - 2015. Испытания нового российского истребителя идут успешно" (https://vpk.name/news/18296_v_interesah_gpv_2015_ispyitaniya_novogo_rossiiskogo_istrebitelya_idut_ushpeshno.html) [Tests of the new Russian fighter are going successfully]. *vpk.name* (in Russian). 7 March 2008. Archived (https://web.archive.org/web/20081005073231/https://vpk.name/news/18296_v_interesah_gpv_2015_ispyitaniya_novogo_rossiiskogo_istrebitelya_idut_ushpeshno.html) from the original on 5 October 2008. Retrieved 6 June 2019.
252. "US Pressured Indonesia to Abandon Su-35 Fighter Jet Contract with Russia" (http://www.defenseworld.net/news/22128/US_Pressured_Indonesia_to_Abandon_Su_35_Fighter_Jet_Contract_with_Russia). *www.defenseworld.net*. Archived (https://web.archive.org/web/20181215174749/http://www.defenseworld.net/news/22128/US_Pressured_Indonesia_to_Abandon_Su_35_Fighter_Jet_Contract_with_Russia) from the original on 15 December 2018. Retrieved 4 January 2019.
253. "Russian Air-To-Air Missile Tests Signal Potential New Capabilities for Flanker and Felon" (<https://www.thedrive.com/the-war-zone/36899/russian-air-to-air-missile-tests-signal-potential-new-capabilities-for-flanker-and-felon>). *TheDrive*. 5 October 2020.

Bibliography

- "Sukhoi fighter updates make debut appearance" (<https://web.archive.org/web/20100706011923/https://www.flightglobal.com/pdfarchive/view/2001/2001%20-%200808.html>). *Flight International*. **159** (4771). London: Reed Business Information: 20. 13–19 March 2001. ISSN 0015-3710 (<https://search.worldcat.org/issn/0015-3710>). Archived from the original (<https://www.flightglobal.com/pdfarchive/view/2001/2001%20-%200808.html>) on 6 July 2010. Retrieved 12 September 2017.
- Andrews, Thomas (Spring 2003). "Su-27/30 family: 'Flanker' in the 21st Century". *International Air Power Review*. Vol. 8. Norwalk, CT: AIRtime Publishing. ISBN 1-880588-54-4.
- Archibald, David (2016). "Resolving the F-35 Nightmare". *Defense & Foreign Affairs Strategic Policy*. **44** (2). Alexandria, VA: International Strategic Studies Association: 12–14. ISSN 0277-4933 (<https://search.worldcat.org/issn/0277-4933>).
- Barrie, Douglas (26 February – 4 March 1997). "Advanced Flanker future hinges on export" (<https://web.archive.org/web/20170807195203/https://www.flightglobal.com/pdfarchive/view/1997/1997%20-%200482.html>). *Flight International*. **151** (4563). London: Reed Business Information: 8. ISSN 0015-3710 (<https://search.worldcat.org/issn/0015-3710>). Archived from the original (<https://www.flightglobal.com/pdfarchive/view/1997/1997%20-%200482.html>) on 7 August 2017.
- ——— (1 September 2003). "Singular Demands". *Aviation Week & Space Technology*. **159** (9). New York: McGraw-Hill: 39. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- Butowski, Piotr (Summer 2004). "Halfway to PAK FA" (<https://web.archive.org/web/20150924122311/http://www.highbeam.com/doc/1G1-126558892.html>). *Interavia Business & Technology* (676). Geneva: Aerospace Media Publishing: 38–41. ISSN 1423-3215 (<https://search.worldcat.org/issn/1423-3215>). Archived from the original (<http://www.highbeam.com/doc/1G1-126558892.html>) on 24 September 2015. Retrieved 19 April 2013.
- ——— (22 September 2007). "Wraps come off new Russian fighters" (<https://web.archive.org/web/20131105233557/http://www.highbeam.com/doc/1G1-170506722.html>). *Interavia Business & Technology* (689). Geneva: Aerospace Media Publishing: 34–36. ISSN 1423-3215 (<https://search.worldcat.org/issn/1423-3215>). Archived from the original (<http://www.highbeam.com/doc/1G1-170506722.html>) on 5 November 2013. Retrieved 19 April 2013.
- Butowski, Piotr (October 2019). "The final Flanker". *Air International*. **97** (4): 36–39. ISSN 0306-5634 (<https://search.worldcat.org/issn/0306-5634>).
- Fink, Donald (6 December 1993). "New Su-35 Boasts Greater Agility". *Aviation Week & Space Technology*. **139** (23). New York: McGraw-Hill: 44–46. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- Fiszer, Michal (August 2004). "Red Fighters Revised". *Journal of Electronic Defense*. **27** (8). Gainesville, FL: Association of Old Crows: 43–44, 46–53. ISSN 0192-429X (<https://search.worldcat.org/issn/0192-429X>).
- Fomin, Andrey (May 2008). "Su-35 Has Flown!" (https://web.archive.org/web/20131029195316/http://en.take-off.ru/pdf_to/to10.pdf) (PDF). *Take-Off*. Moscow: Aeromedia Publishing: 24–29. Archived from the original (http://en.take-off.ru/pdf_to/to10.pdf) (PDF) on 29 October 2013. Retrieved 26 October 2013.
- ——— (July 2012). "Su-35S in trials" (https://web.archive.org/web/20131029193146/http://en.take-off.ru/pdf_to/to23.pdf) (PDF). *Take-Off*. Moscow: Aeromedia Publishing: 16–17. Archived from the original (http://en.take-off.ru/pdf_to/to23.pdf) (PDF) on 29 October 2013. Retrieved 26 October 2013.
- ——— (July 2014). "Su-35S Being Fielded With Success" (https://web.archive.org/web/20170602213831/http://en.take-off.ru/pdf_to/to29.pdf) (PDF). *Take-Off*. Moscow: Aeromedia

Publishing: 36–39. Archived from the original (http://en.take-off.ru/pdf_to/to29.pdf) (PDF) on 2 June 2017.

- Gethin, Howard (9–15 September 1998). "Sukhoi flies latest Su-37 demonstrator" (<https://web.archive.org/web/20121108112022/http://www.flightglobal.com/pdfarchive/view/1998/1998%20-%202468.html>). *Flight International*. **154** (4642). London: Reed Business Information: 32. ISSN 0015-3710 (<https://search.worldcat.org/issn/0015-3710>). Archived from the original (<http://www.flightglobal.com/pdfarchive/view/1998/1998%20-%202468.html>) on 8 November 2012. Retrieved 26 October 2013.
- Gill, Bates; Kim, Taeho (1995). *China's Arms Acquisitions from Abroad: A Quest for 'Superb and Secret Weapons'* (<https://archive.org/details/chinasarmsacquis0000gill>). Oxford, UK: Oxford University Press. ISBN 978-0-19-829196-1.
- Gordon, Yefim (2007). *Sukhoi Su-27. Famous Russian Aircraft*. Hinckley, UK: Midlands Publishing. ISBN 978-1-85780-247-4.
- Joo, Seung-Ho; Kwak, Tae-Hwan Kwak (2001). *Korea in the 21st Century*. Hauppauge, NY: Nova Publishers. ISBN 978-1-56072-990-7.
- Komarov, Alexey (4 February 2002). "Brazil Inks Fighter and Launcher Pacts with Russia and Ukraine". *Aviation Week & Space Technology*. **156** (5). New York: McGraw-Hill: 61. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- Kopp, Carlo (March 2010). "Sukhoi's Su-35S – not your father's Flanker". *Defence Today*. Amberley, Queensland: Strike Publications: 8–10. ISSN 1447-0446 (<https://search.worldcat.org/issn/1447-0446>).
- Mader, Georg (March 2007). "Los FLANKERs del Presidente". *Military Technology*. **31** (3). Bonn: Monch Publications: 102–104. ISSN 0722-3226 (<https://search.worldcat.org/issn/0722-3226>).
- Novichkov, Nicolay (26 August 1996). "Sukhoi Set to Exploit Thrust Vector Control". *Aviation Week & Space Technology*. **145** (9). New York: McGraw-Hill: 52. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- Pyadushkin, Maxim (13 December 2010). "Russian Revival". *Aviation Week & Space Technology*. **172** (45). New York: McGraw-Hill: 28. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- ———; Perrett, Bradley (27 October 2014). "Back for More". *Aviation Week & Space Technology*. **176** (38). New York: Penton Media: 18–19. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>).
- Rivers, Brendan (February 2005). "Brazil Delays Fighter Program Again". *Journal of Electronic Defense*. **28** (2). Gainesville, Florida: Association of Old Crows: 18. ISSN 0192-429X (<https://search.worldcat.org/issn/0192-429X>).
- Sherman, Kenneth B. (1 September 2001). "US to ROK: "Buy American"" (<https://web.archive.org/web/20131105233338/http://www.highbeam.com/doc/1G1-78639426.html>). *Journal of Electronic Defense*. **24** (9). Gainesville, Florida: Association of Old Crows: 22. ISSN 0192-429X (<https://search.worldcat.org/issn/0192-429X>). Archived from the original (<http://www.highbeam.com/doc/1G1-78639426.html>) on 5 November 2013. Retrieved 22 April 2013.
- Sweetman, Bill (April 2004). "Russian stealth research revealed" (<https://web.archive.org/web/20131105233528/http://www.highbeam.com/doc/1G1-126317422.html>). *Journal of Electronic Defense*. **27** (1). Gainesville, Florida: Association of Old Crows: 26–27. ISSN 0192-429X (<https://search.worldcat.org/issn/0192-429X>). Archived from the original (<http://www.highbeam.com/doc/1G1-126317422.html>) on 5 November 2013. Retrieved 22 April 2013.
- ——— (24 June 2013). "Tight Corners" (<http://aviationweek.com/awin/sukhoi-test-pilot-explains-supermaneuverability>). *Aviation Week & Space Technology*. **175** (21). New York: Penton Media: 33. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>). Retrieved 4 August 2017.

- ——— (19 August 2013). "New Moves" (<http://aviationweek.com/awin/sukhoi-t-50-shows-flight-control-innovations>). *Aviation Week & Space Technology*. **175** (28). New York: Penton Media: 43–44. ISSN 0005-2175 (<https://search.worldcat.org/issn/0005-2175>). Retrieved 23 July 2017.
- Velovich, Alexander (8–14 May 1996). "Thrust-vectoring Sukhoi Su-35 flies" (<https://web.archive.org/web/20110314022230/http://www.flightglobal.com/pdfarchive/view/1996/1996%20-%201112.html>). *Flight International*. **149** (4522). London: Reed Business Publishing: 16. ISSN 0015-3710 (<https://search.worldcat.org/issn/0015-3710>). Archived from the original (<https://www.flightglobal.com/pdfarchive/view/1996/1996%20-%201112.html>) on 14 March 2011. Retrieved 24 July 2017.
- Williams, Mel, ed. (2002). "Sukhoi 'Super Flankers' ". *Superfighters: The Next Generation of Combat Aircraft*. Norwalk, Connecticut: AIRtime Publishing. ISBN 1-880588-53-6.
- Zhao, Suisheng (2004). *Chinese Foreign Policy: Pragmatism and Strategic Behavior*. Armonk, New York: M. E. Sharpe. ISBN 978-0-7656-1284-7.

External links

- KnAAPO (<http://www.knaapo.ru/products/su-35/>)
- Rosoboronexport (<http://roe.ru/eng/catalog/aerospace-systems/fighters/su-35/presentation#1>)

External videos

on *History Channel* Official YouTube Channel(in Russian)

 "The best Sukhoi aircraft". Su-35S (<https://www.youtube.com/watch?v=FdhVQ8nnu1g>) on YouTube

Retrieved from "https://en.wikipedia.org/w/index.php?title=Sukhoi_Su-35&oldid=1323927832"